

**A WEB-BASED MEDICAL APPOINTMENT AND SCHEDULING SYSTEM FOR
THE MINDALANO SPECIALIST HOSPITAL FOUNDATION INC.**

COLLEGE OF INFORMATION AND COMPUTING STUDIES
Mindanao State University – Main Campus

A Capstone Project 1
Presented to the Faculty of
College of Information and Computing Sciences

In Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Information Technology

By:
Ibrahim, Eaz Tharry
H. Jalal, Saripoden D.
Nasroden, Aynoden A.

Azimah D. Ampuan
Adviser

Amer Hussien T. Macatotong
Co-Adviser

December 2025

TABLE OF CONTENTS

Chapter 1: INTRODUCTION

Introduction	1
1.1. Project Context	2
1.2. Purpose and Description	3
1.3. Objectives	4
1.3.1. General Objective	4
1.3.2. Specific Objectives	4
1.4. Scope and Limitation	5
1.5. Significance of the Study	6
1.6. Definition of Terms	6

Chapter 2: REVIEW OF RELATED LITERATURE

2.1. Review of the Related Concept	
2.1.1. Health Information Systems (HIS)	9
2.1.2. Web-Based Appointment Scheduling Systems	10
2.1.3. Patient-Centered Care in Digital Health	11
2.1.4. Role-Based Access Control (RBAC) in Healthcare Systems	12
2.1.5. Automated Notification Systems in Healthcare	13
2.1.6. Electronic Medical Record (EMR)	14
2.1.7. Lanao del Sur: Local Healthcare Context	15
2.2. Review of Related Literature	
2.2.1. Automated Notification and Reminder Systems in Healthcare	17
2.2.2. Automated Patient Self-Scheduling in Healthcare Systems	17
2.2.3. Effectiveness of Internet-Based Appointment Systems in Healthcare	18
2.2.4. Electronic Medical Records Adoption in Community and Outpatient Healthcare	19
2.2.5. Patient-Centric Web-Based Appointment Scheduling Systems	19
2.3. Summary of Related Literature	20
2.4. Review of Related Systems	20

2.4.1. A Web-Based Doctor Appointment System	21
2.4.2. A Web-Based Online Doctor's Appointment and Medical Database Management System.....	22
2.4.3. PEC e-SUS APS Online Appointment Scheduling System.....	23
2.4.4. Smart Web-Based Medical Appointment Booking System	25
2.5. Summary of the Related System.....	26
2.6. Technical Background	27
2.6.1. Organizational Chart	29
2.6.2. Workflow	
2.6.2.1. Current Workflow	31
2.6.2.2. Proposed Workflow	33

Chapter 3: METHODOLOGY

3.1. Scrum Framework	36
3.2. Requirement Analysis.....	37
3.2.1. Empathy Map	38
3.2.2. Functional And Non-Functional Requirement	41
3.2.3. Gantt Chart	42

3.3. DESIGN

3.3.1. Hierarchical Input-Process Output	44
3.3.2. Input Process-Output Diagram	45
3.3.2.1. Patient Input-Process-Output	45
3.3.2.2. Secretary Input-Process-Output	47
3.3.2.3. Doctor Input-Process-Output	48
3.3.2.4. Admin Input-Process-Output	49
3.3.3. Use Case Diagram	51
3.3.4. Entity-Relationship Diagram.....	52
3.3.5. Architectural Design.....	53

3.4. DEVELOPMENT

3.4.1. Software Requirements	54
3.4.2. Hardware Requirements	56
3.5. Testing.....	58
3.5.1. Test Plan	58
3.5.2. Respondents.....	60

3.5.3. Tools (Questionnaire)	61
REFERENCES.....	63

LIST OF FIGURES

Figure 2.1: Mindalano Specialist Hospital Foundation, Inc.	16
Figure 2.2: A Web-Based Doctor Appointment System	21
Figure 2.3: A Web-Based Online Doctor's Appointment and Medical Database Management System.....	22
Figure 2.4: PEC e-SUS Aps Online Appointment Scheduling System	24
Figure 2.5: Smart Web-Based Medical Appointment Booking System.....	25
Figure 2.6: Organizational Chart	30
Figure 2.7: Current Workflow of MSHFI	32
Figure 2.8: Proposed Workflow of MSHFI	33
Figure 3.1: Agile Methodology	35
Figure 3.2: Scrum Framework	36
Figure 3.3: Empathy Map of Patient.....	38
Figure 3.4: Empathy Map of Doctor.....	39
Figure 3.5: Empathy Map of Secretary	40
Figure 3.6: MSHFI Web-based Hierarchical Input-Process-Output	44
Figure 3.7: Input-Process-Output for User Account	45
Figure 3.8: Input-Process-Output for Booking Appointment	46
Figure 3.9: Input-Process-Output for Feedback.....	46
Figure 3.10: Input-Process-Output for Confirming Appointment	47
Figure 3.11: Input-Process-Output for View Medical Records	47
Figure 3.12: Input-Process-Output for User Account	48
Figure 3.13: Input-Process-Output for confirming/rescheduling an Appointment.....	48
Figure 3.14: Input-Process-Output for viewing medical Records for a patient.....	49
Figure 3.15: Input-Process-Output for User Account	49
Figure 3.16: Input-Process-Output for Manage User	50
Figure 3.17: Input-Process-Output for Feedback & Logs Review	50
Figure 3.18: Use Case Diagram	51
Figure 3.19: Entity-Relationship Diagram.....	52
Figure 3.20: Architectural Design.....	53

LIST OF TABLES

Table 2.1: Summary of Related Systems	27
Table 3.1: Functional and Non-Functional Requirements	41
Table 3.2: Gantt Chart.....	43
Table 3.3: Developer’s Software Requirements	54
Table 3.4: User’s Software Requirements.....	54
Table 3.5: Admin’s Software Requirements	55
Table 3.6: Server’s Software Requirements.....	55
Table 3.7: Developer’s Hardware Requirements	56
Table 3.8: User’s Hardware Requirements	56
Table 3.9: Admin’s Hardware Requirements	57
Table 3.10: Server’s Hardware Requirements	58
Table 3.11: System Usability Scale (SUS) Questionnaire	61

CHAPTER 1

Introduction

Healthcare information systems are recognized as essential innovations in improving service delivery and efficiency in medical institutions. According to the World Health Organization (2022), a well-implemented health information system plays a critical role in strengthening healthcare accessibility, reducing delays, and ensuring proper management of patient data. Appointment and scheduling systems, in particular, are vital in minimizing waiting times, avoiding schedule conflicts, and providing patients with timely medical consultations. Locally, De Guzman (2021) defines hospital scheduling systems as digital platforms designed to assist patients, doctors, and staff in managing appointments and records, thereby transforming traditional manual workflows into automated processes.

Despite the benefits of modern scheduling systems, many healthcare institutions in the Philippines still rely on manual processes. This results in long queues, overlapping appointments, misplaced records, and delays in consultations. These challenges are also experienced at Mindalano Specialist Hospital Foundation Inc., where appointment scheduling, patient registration, and record-keeping are all done manually. Patient records are stored on paper, which makes retrieval slow and prone to loss. Queue numbers are issued, but patients, especially the elderly, often wait for long periods without knowing when they will be called. The hospital also has a limited number of physicians, and sudden doctor absences sometimes lead to unannounced appointment cancellations. As a result, patients may travel long distances only to discover that consultations are unavailable, causing inconvenience and dissatisfaction.

To address these challenges, the proponents propose the development of a Web-Based Medical Appointment and Scheduling System for the Mindalano Specialist Hospital Foundation Inc. The system will allow patients to book appointments with their preferred

doctor, check doctor availability, and receive notifications in case of cancellations or schedule changes. It will also include digital patient records for easier retrieval and role-based access for doctors, staff, and patients. With this system, patients especially the elderly will no longer have to wait in uncertainty, hospital staff will have a streamlined workflow, and doctors will be able to manage their schedules more efficiently.

1.1. Project Context

The project focuses on the digital transformation of the Mindalano Specialist Hospital Foundation Inc.'s manual processes related to appointment booking, patient registration, and medical record management. The hospital currently relies on paper-based documentation and walk-in scheduling, which results in long waiting times, misplaced records, scheduling conflicts, and inefficiencies in daily operations. The proposed system seeks to address these issues by converting the hospital's existing manual workflow into an organized and automated digital process.

The project follows a digital transformation and process automation approach, wherein traditional hospital procedures are translated into an electronic system that centralizes data and streamlines transactions. Instead of manually recording patient information and appointments, the system will automate scheduling, record storage, and appointment monitoring through a single online platform. This approach allows real-time access to information, reduces repetitive administrative tasks, and improves coordination among patients, doctors, and hospital staff.

The proposed solution will be developed as a web-based medical appointment and scheduling system built using the Django web framework. The system will adopt a client-server architecture, where users access the platform through standard web browsers on desktop or mobile devices, while all application logic and data processing

are handled on the server. This architecture supports centralized data management, controlled access, and efficient communication between system components.

Front-end web technologies will be used to provide a simple and user-friendly interface designed to accommodate all users, including elderly patients. A relational database will be used to securely store patient profiles, appointment records, doctor schedules, and medical information. To further support efficient hospital operations, the system will include automated email notifications for appointment confirmations, reminders, and schedule changes, as well as role-based access control to ensure that system features are accessible only to authorized users.

Through this high-level transformation approach and the use of web-based technologies, the project aims to modernize hospital appointment management, improve workflow efficiency, and enhance patient experience by providing a reliable and accessible digital scheduling solution tailored to the needs of the Mindalano Specialist Hospital Foundation Inc.

1.2. Purpose and Description

This project aims to design and develop a Web-Based System that will address the inefficiencies caused by the hospital's current manual system of patient registration, appointment scheduling, and medical record management. The proposed system seeks to modernize the hospital's operations by providing a digital platform that will streamline processes, minimize errors, and improve healthcare service delivery.

This system is intended to allow patients to book, reschedule, or cancel appointments with their preferred doctors and to check real-time doctor availability. It will also include a centralized medical record database to ensure that patient histories

are properly stored and easily retrievable by hospital staff. In addition, automated notifications via email will be integrated to remind patients of their upcoming appointments or inform them of sudden schedule changes. By implementing this system, the hospital will be able to reduce long waiting times, manage doctor schedules effectively, and ensure better coordination between patients, staff, and physicians.

The project officially started in **October 2025** and is expected to be completed by the year **2026**. The major participants in the development of this project are the proponents from the Bachelor of Science in Information Technology program, in collaboration with the doctors, staff and administrators of the Mindalano Specialist Hospital Foundation Inc. Through this project, the proponents aim to enhance the hospital's service delivery, including its appointment and scheduling system, and contribute to the advancement of healthcare technology in the local community.

1.3. Objectives

1.3.1. General Objective

To develop a web-based medical appointment and scheduling system for the Mindalano Specialist Hospital Foundation Inc. to streamline hospital operations, improve service efficiency, and enhance patient satisfaction.

1.3.2. Specific Objectives

1. To gather and analyze information on the hospital's current appointment scheduling, patient registration, and medical record management processes to identify existing problems, workflow gaps, and system requirements that will serve as the basis for system development.
2. To design a structured relational database that will securely store and organize patient information, doctor schedules, appointment records, and

medical data, ensuring accurate data retrieval, reduced redundancy, and efficient system operation.

3. To develop a Web-Based Medical Appointment and Scheduling System using the Django framework that includes core components such as user authentication, appointment booking and rescheduling, doctor availability management, electronic medical records, role-based access control, and automated email notifications for patients, doctors, and hospital staff.
4. To test and evaluate the functionality, performance, and usability of the developed system using the System Usability Scale (SUS) to determine its effectiveness, reliability, and user acceptance.

1.4. Scope and Limitations

The system is limited to scheduled outpatient consultations and does not support emergency cases, inpatient services, or unscheduled walk-in consultations, as these require immediate medical attention and separate clinical workflows.

The system operates as a web-based platform and therefore requires a stable internet connection for access and real-time updates. System availability and performance are constrained by server capacity, network reliability, and hardware resources provided by the hospital. In areas with limited internet connectivity, system access may be delayed or interrupted.

The project does not include mobile application integration for Android or iOS platforms and does not support advanced features such as predictive analytics, telemedicine, or integration with external hospital information systems in its initial version. Development is also constrained by time and resource limitations, as the project is implemented within the academic timeframe and scope of a capstone project.

Additionally, system effectiveness depends on user readiness and proper system adoption. Hospital staff and patients with limited digital literacy may require orientation or assistance to fully utilize the system. The electronic medical record component is limited to basic outpatient information and does not replace a comprehensive hospital-wide EMR system.

1.5. Significance of the Study

Patients – They will benefit from a more organized and efficient appointment process, shorter waiting times, and convenient access to schedules and notifications.

Doctors – They will gain better control over their schedules, minimize overlapping appointments, and provide consultations in a more organized manner.

Hospital Staff – Their workload will be reduced with the automation of patient registration, record retrieval, and schedule management.

Mindalano Specialist Hospital Foundation Inc. – The institution will be able to modernize its services, improve patient satisfaction, and strengthen its operational efficiency.

Future Researchers – This project can serve as a valuable reference for further research and development of hospital information systems and related technologies.

1.6. Definition of Terms

This section provides the definitions of key terms used throughout the study to ensure clarity and consistency in understanding the concepts related to the proposed system.

Appointment – a scheduled consultation between a patient and a doctor at a specific date and time. This term is significant because the core function of the proposed system

is to automate appointment setting to prevent long waiting times and scheduling conflicts.

Web-Based System – a software application that runs on a web browser and can be accessed through the internet without requiring installation. This is relevant to the study because the proposed system relies on web access to allow patients and staff to use it conveniently from different devices.

Scheduling – the process of organizing and assigning time slots for patient consultations with available doctors. This is important because the system aims to improve the hospital's current manual scheduling process and enhance workflow efficiency.

Electronic Medical Record (EMR) – a digital version of a patient's medical history, including personal information, past consultations, and other relevant health data. This is significant because it supports faster retrieval of patient information compared to the hospital's manual record system.

Patients – Individuals who seek medical consultations and use the system to book, manage, or cancel appointments. They are essential to the study because they represent the main users who will benefit from reduced waiting time and improved convenience.

Doctors – licensed medical professionals who provide consultations and manage their availability within the system. They are important to the system because their schedules and availability determine how appointments are arranged and confirmed.

Secretary – hospital personnel responsible for assisting patients, managing appointment records, and supporting daily administrative operations. Their role is significant because the system helps reduce their workload and errors caused by manual processes.

Administrator (Admin) – a user with the highest level of system access, responsible for managing user accounts, schedules, system settings, and records. This term is important because admins ensure that the system functions properly and that users have the correct access.

Queue Number – a number assigned to a patient to determine their order of consultation. This is significant because the current manual queueing system is one of the major problems the project aims to improve.

User Interface (UI) – the visual part of the system that users interact with, including forms, buttons, menus, and pages. This is important because a simple and clear interface ensures that even elderly patients can use the system easily.

Database – a structured system for storing and organizing information such as patient records, schedules, and appointment details. This is significant because the proposed system depends on a reliable database to securely manage all medical and scheduling information.

Role-Based Access Control – a security mechanism that restricts system features based on user roles such as patient, doctor, secretary, or admin. This is important because it protects sensitive data and ensures that users only access functions appropriate to their role.

Manual System – a traditional process where tasks such as scheduling, record-keeping, and registration are performed by hand without the use of digital tools. This term is relevant because the study aims to replace the hospital's existing manual system with a more efficient automated solution.

CHAPTER 2

Review of Related Literature and Systems

This chapter discusses relevant literature and existing systems that support the development of A Web-Based Medical Appointment and Scheduling System for the Mindalano Specialist Hospital Foundation Inc. It focuses on scientific and technological concepts related to health information systems, digital appointment management, patient-centered scheduling, and data security in healthcare. It also includes different studies and systems that are affiliated with the project that the proponents are going to develop.

2.1. Review of Related Concepts

2.1.1. Health Information Systems (HIS)

A Health Information System (HIS) is a digital infrastructure designed to collect, store, manage, and transmit health-related data to support clinical, administrative, and decision-making processes in healthcare institutions. According to the World Health Organization (2022), a well-implemented HIS strengthens healthcare accessibility, reduces delays in service delivery, and ensures the accuracy and confidentiality of patient records.

In hospital settings, HIS components include electronic medical records, registration modules, scheduling tools, and reporting dashboards. These systems transform traditional paper-based workflows into automated, real-time digital processes enhancing both operational efficiency and patient experience.

For this project, the proposed system functions as a specialized module of a Health Information System, focusing specifically on appointment scheduling and basic electronic medical record management. The system will be developed using the Django web framework, which will handle server-side processing, user

authentication, appointment workflows, and secure data management. Django is suitable for healthcare systems due to its structured architecture, built-in security features, and scalability.

The system will utilize a relational database management system to store patient profiles, appointment records, doctor schedules, and medical information in an organized and centralized manner. Through a web-based platform, authorized users will be able to access real-time scheduling data and patient records, enabling faster service delivery and improved coordination. By digitizing appointment workflows using Django-based technologies, the system directly addresses the hospital's challenges related to long queues, manual record handling, and lack of real-time updates.

2.1.2. Web-Based Appointment Scheduling Systems

A web-based appointment scheduling system is a digital platform that allows patients to view doctor availability, book consultations, and receive automated confirmations or updates through a web browser. Unlike manual scheduling where appointments are managed via phone calls, paper logs, or walk-in queues web-based systems provide transparency, real-time synchronization, and self-service capabilities.

Effective scheduling systems have several key features that improve efficiency and user experience. They provide real-time calendar views of doctor availability, allowing patients and staff to easily see open time slots and avoid scheduling conflicts. These systems also include online registration and patient profile creation, making it convenient for patients to manage their information and appointments digitally. Additionally, automated email notifications are sent to confirm bookings, remind patients of upcoming appointments, or inform them of

cancellations. Finally, role-based dashboards are available for patients, doctors, and staff, ensuring that each user can access relevant information and tools suited to their specific needs.

In the context of Mindalano Specialist Hospital, where patients often travel long distances and elderly clients struggle with uncertain waiting times, such a system ensures predictability and convenience. The proposed platform will allow patients to secure a time slot in advance, eliminating the need to wait indefinitely in the hospital lobby. Moreover, in cases where a doctor has an emergency and cannot attend, the system will immediately notify affected patients, preventing wasted trips and reducing frustration.

2.1.3. Patient-Centered Care in Digital Health

Patient-centered care is a healthcare approach that prioritizes the needs, preferences, and values of patients in every aspect of service delivery. In digital health, this principle translates into systems that are accessible, easy to use, and designed with the end-user in mind especially for vulnerable groups such as the elderly or those with limited digital literacy.

The World Health Organization (2022) emphasizes that digital health tools must be inclusive and culturally appropriate to ensure equitable access. For hospitals in Mindanao, this means designing interfaces that are simple, visual, and available in local languages (e.g., Filipino, Cebuano, or Maranao).

The proposed system embodies a patient-centered design by focusing on simplicity, accessibility, and convenience. It allows users to book an appointment in just a few easy steps like entering their name, selecting a doctor, choosing a date and time, and confirming the booking. The interface features large buttons and clear labels to assist elderly users and those who may struggle with complex

digital systems. Additionally, the system uses email notifications, a widely accessible communication channel in the Philippines, rather than relying solely on email or app alerts. It also includes digital patient profiles, which eliminate the need for repetitive registration and help maintain continuity of care by keeping patient information organized and easily accessible.

By placing the patient at the center of the design process, the system not only improves usability but also builds trust and satisfaction that is critical for a community hospital serving a loyal but underserved clientele.

2.1.4. Role-Based Access Control (RBAC) in Healthcare Systems

Role-Based Access Control (RBAC) is a security model that restricts system access based on user roles to protect sensitive health information. In a medical scheduling system, Role-Based Access Control (RBAC) ensures that each user can only access functions appropriate to their role, maintaining both security and efficiency. Patients are limited to viewing and managing their own appointments, preventing unauthorized access to others' information. Doctors, on the other hand, can access their schedules, view assigned patient records, and update their availability as needed. Meanwhile, hospital staff have permissions to register patients, monitor the appointment queue, and retrieve medical histories, but they are restricted from modifying doctor schedules. This structured access control helps protect sensitive data and ensures that system operations remain well-organized and secure.

This model aligns with the Data Privacy Act of 2012 (Republic Act No. 10173) and the Philippine Health Data Privacy Guidelines, which require strict safeguards for personal health information. The system will implement RBAC using J session management and middleware authentication, ensuring that data is

only visible to authorized personnel. This protects both the hospital and its patients from unauthorized access especially important in a small institution where staff handle multiple roles.

2.1.5. Automated Notification Systems in Healthcare

Automated notification systems use email or in-app alerts to inform patients about appointment confirmations, reminders, or last-minute changes. In the Philippine healthcare landscape, digital communication channels are increasingly vital for timely and efficient patient engagement, particularly in facilities serving both urban and semi-urban communities with growing internet access.

For Mindalano Specialist Hospital, where doctor emergencies often lead to unannounced cancellations, an automated email notification feature is essential. The system will integrate with secure email services to send real-time messages such as:

“Subject: Urgent: Your Appointment Has Been Cancelled

Dear [Patient Name],

Your appointment with Dr. Mindalano on November 18 at 10:00 AM has been cancelled due to an emergency. We sincerely apologize for the inconvenience.

Please log in to your patient portal to reschedule at your earliest convenience.

Thank you for your understanding.

— Mindalano Specialist Hospital”

This functionality directly responds to the problem stated in Chapter I, where patients are left waiting with no updates, thereby saving time, reducing confusion, and improving care coordination.

2.1.6. Electronic Medical Record (EMR)

An Electronic Medical Record (EMR) is a digital version of a patient's paper chart, containing comprehensive health information such as personal details, medical history, diagnoses, medications, treatment plans, immunization dates, allergies, and laboratory test results. Unlike traditional filing systems, EMRs allow authorized healthcare providers to instantly access, update, and share patient data across departments, improving clinical decision-making and continuity of care.

According to the World Health Organization (2022), EMRs are a foundational component of modern Health Information Systems, particularly in outpatient settings where patients return for follow-up consultations. They reduce the risk of lost files, duplicate tests, and transcription errors, while also supporting faster service delivery critical in busy clinics like Mindalano Specialist Hospital Foundation Inc.

In the current manual setup at the hospital, staff must physically search through folders to retrieve past records which is a process that is not only time-consuming but also prone to misfiling or damage, especially during peak hours. Elderly patients with chronic conditions (e.g., hypertension or diabetes) often require access to their previous lab results or prescriptions, yet delays in record retrieval can disrupt consultation flow and compromise care quality.

The proposed system will integrate a simplified EMR module that stores essential patient data digitally. Each patient will have a unique profile linked to their appointment history, allowing doctors to view past consultations and prescribed medications, track recurring health issues over time, and avoid

redundant questioning during check-ups, thereby enabling more efficient and informed clinical decision-making.

The EMR will be built using MySQL relational database structures, ensuring data integrity and efficient querying. Access will be protected through role-based permissions where only assigned doctors and authorized staff can view or edit medical entries to ensure compliance with the Data Privacy Act of 2012.

While the system will not replace a full-featured hospital EMR (such as those used in large tertiary hospitals), it provides a practical, scaled-down solution tailored to the outpatient-focused operations of Mindalano Specialist Hospital. This approach aligns with recommendations from the Department of Health (2024), which encourages community hospitals to adopt modular digital tools that address specific workflow gaps rather than implementing complex, costly systems all at once.

By embedding EMR functionality into the appointment system, the project ensures that scheduling and medical history are interconnected, creating a more holistic and efficient patient care experience.

2.1.7. Lanao del Sur: Local Healthcare Context

Lanao del Sur is a landlocked province in the Bangsamoro Autonomous Region in Muslim Mindanao (BARMM), known for its rich Maranao culture, historical significance, and scenic landscapes surrounding Lake Lanao. With a population exceeding 1.2 million (PSA, 2020), the province faces persistent challenges in healthcare access, including limited specialist services, geographic isolation of barangays, and under-resourced medical facilities.

The capital city, Marawi, and its surrounding municipalities rely heavily on a few private and public hospitals for secondary care. Mindalano Specialist

Hospital Foundation Inc., though small in scale, plays a critical role in providing outpatient and consultation services to residents from Marawi and nearby towns. Many patients travel by foot, motorcycle, or public utility vehicle for hours to reach the hospital making predictable appointment times not just a convenience, but a necessity.

Moreover, Lanao del Sur has a significant elderly population, many of whom manage chronic conditions such as hypertension and diabetes. For these patients, repeated trips due to missed or unannounced cancellations can be physically and financially draining. The lack of digital infrastructure in many barangays further complicates communication, reinforcing the need for low-tech-compatible solutions like email-based notifications.

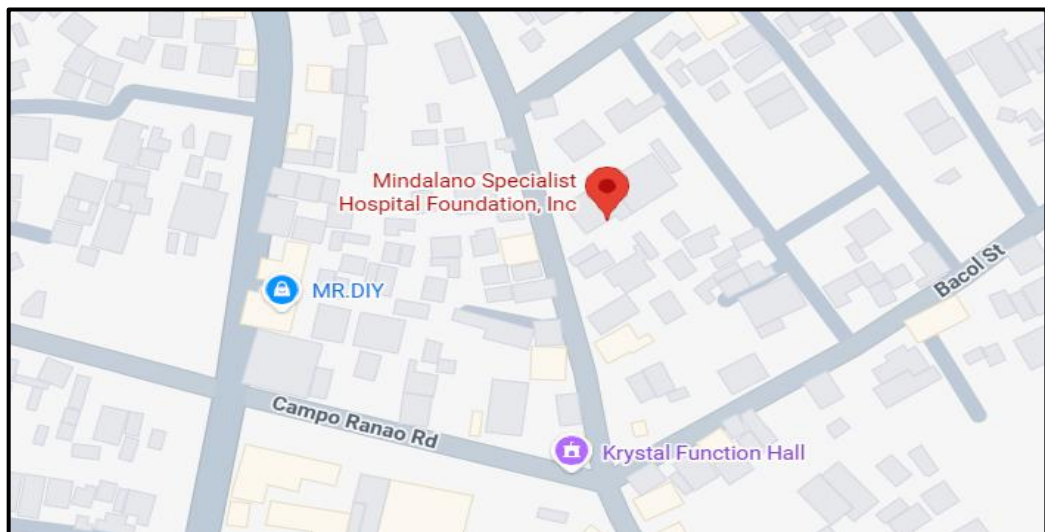


Figure 2.1 Mindalano Specialist Hospital Foundation, Inc.

This project is rooted in this local reality. By developing a system that is responsive to the socio-geographic context of Lanao del Sur, the proponents aim not only to modernize one hospital but also to create a replicable model for community-based healthcare digitization in underserved Mindanao provinces.

2.2. Review of Related Literature

2.2.1. Automated Notification and Reminder Systems in Healthcare

Aljabri et al. (2023) investigated the impact of automated notification and reminder systems on patient engagement and appointment compliance in healthcare settings. The study demonstrated that automated reminders significantly reduce no-show rates and improve patient adherence to scheduled consultations. Timely notifications regarding appointment confirmations, cancellations, or rescheduling were found to enhance patient trust and reduce frustration, particularly in community-based hospitals where patients often travel long distances. The researchers emphasized that automated notification systems are essential for improving communication and coordination between healthcare providers and patients.

This study directly addresses the challenges faced by Mindalano Specialist Hospital Foundation Inc., where patients are often uninformed about sudden changes in doctor availability. Incorporating automated notifications into the proposed system ensures timely communication and reduces unnecessary waiting and travel.

2.2.2. Automated Patient Self-Scheduling in Healthcare Systems

Lin et al. (2022) analyzed the implementation of automated patient self-scheduling systems in healthcare facilities and their effects on operational efficiency. The study found that self-scheduling platforms significantly reduce missed appointments and scheduling conflicts while decreasing the workload of hospital staff by minimizing manual booking processes. Patients benefited from greater flexibility and control over appointment selection, leading to increased satisfaction and engagement. The researchers concluded that automated self-

scheduling systems contribute to improved service efficiency and better coordination between patients and healthcare providers.

This literature is relevant to the proposed system, as Mindalano Specialist Hospital Foundation Inc. currently relies on manual appointment coordination. Integrating automated self-scheduling functionality will help streamline appointment management and improve both patient convenience and hospital workflow.

2.2.3. Effectiveness of Internet-Based Appointment Systems in Healthcare

A recent study published in *Medicine* (2023) compared internet-based appointment systems with traditional manual appointment methods in hospital settings. The study found that internet-based appointment systems significantly reduce patient waiting time, improve appointment efficiency, and increase overall patient satisfaction. Results showed that patients using online appointment platforms experienced smoother scheduling processes and better time management compared to those using walk-in or phone-based booking. The researchers concluded that internet-based appointment systems offer measurable advantages in improving healthcare service quality and operational efficiency.

This literature further strengthens the proposed project by providing empirical evidence that digital appointment systems are more effective than manual scheduling methods. Implementing a web-based appointment system at Mindalano Specialist Hospital Foundation Inc. is therefore supported by proven improvements in efficiency and patient satisfaction.

2.2.4. Electronic Medical Records Adoption in Community and Outpatient

Healthcare

Epizitone (2023) conducted a systematic review on the adoption and impact of Electronic Medical Records in healthcare institutions, particularly in community and outpatient settings. The study found that EMR systems significantly improve the accessibility, accuracy, and continuity of patient information by replacing paper-based records with centralized digital databases. The review emphasized that EMRs reduce record loss, minimize documentation errors, and support faster clinical decision-making, especially for patients with recurring consultations or chronic conditions. It also noted that successful EMR implementation in smaller healthcare facilities requires simplified and workflow-oriented system designs rather than complex enterprise solutions.

This literature is directly relevant to the proposed Web-Based Medical Appointment and Scheduling System for Mindalano Specialist Hospital Foundation Inc., where patient records are currently managed manually. Integrating a simplified EMR module will enable authorized doctors and staff to retrieve patient histories efficiently, support continuity of care, and reduce delays during consultations.

2.2.5. Patient-Centric Web-Based Appointment Scheduling Systems

Chung (2020) examined the implementation of patient-centric scheduling systems supported by health information technology to improve access to healthcare services and patient experience. The study revealed that web-based appointment scheduling enables patients to independently book consultations, choose preferred healthcare providers, and receive automated confirmations, which significantly reduces waiting times and administrative workload. The

findings showed improved appointment adherence and reduced no-show rates, particularly in outpatient healthcare facilities where appointment demand is high. The study emphasized that patient-centered scheduling enhances healthcare accessibility while improving workflow efficiency for healthcare providers.

This literature supports the proposed Web-Based Medical Appointment and Scheduling System for Mindalano Specialist Hospital Foundation Inc., as it highlights the importance of allowing patients to manage appointments conveniently and reducing uncertainty in consultation schedules.

2.3. Summary of the Related Literature

The reviewed literature indicates that patient-centric web-based appointment scheduling systems, automated self-scheduling platforms, electronic medical records, and notification technologies significantly improve healthcare service delivery. Studies show that digital appointment management reduces waiting times, minimizes scheduling conflicts, and enhances access to care. Automated self-scheduling systems improve hospital efficiency by reducing staff workload, while notification systems decrease missed appointments and improve patient satisfaction. Recent evidence also confirms that internet-based appointment systems are more effective than traditional manual methods in improving efficiency and service quality. Furthermore, literature on Electronic Medical Records highlights their role in ensuring accurate, accessible, and continuous patient care, particularly in community and outpatient healthcare settings. Collectively, these studies provide strong justification for the development of a Web-Based Medical Appointment and Scheduling System for Mindalano Specialist Hospital Foundation Inc., as digital solutions effectively address the limitations of manual scheduling and record-keeping while improving efficiency, coordination, and overall patient experience.

2.4. Review of Related Systems

2.4.1. A Web-Based Doctor Appointment System

This study focuses on the development of a web-based doctor appointment system designed to enhance the efficiency of healthcare delivery by automating the manual appointment process. This was implemented using PHP, Apache, HTML, CSS, and MySQL technologies. Its main goal is to provide a user-friendly, secure, and cost-effective platform that allows patients to register, schedule appointments, track medical histories, and manage prescriptions online. This adopts a multi-layer architecture composed of a user interface, business logic, and database storage. It supports operations for different users, including patients, doctors, and administrators. The admin can manage doctor profiles, schedule new sessions, and oversee patient bookings, while doctors can view appointments, manage their schedules, and update patient information. Patients can register, view available doctors, and book/cancel appointments conveniently.

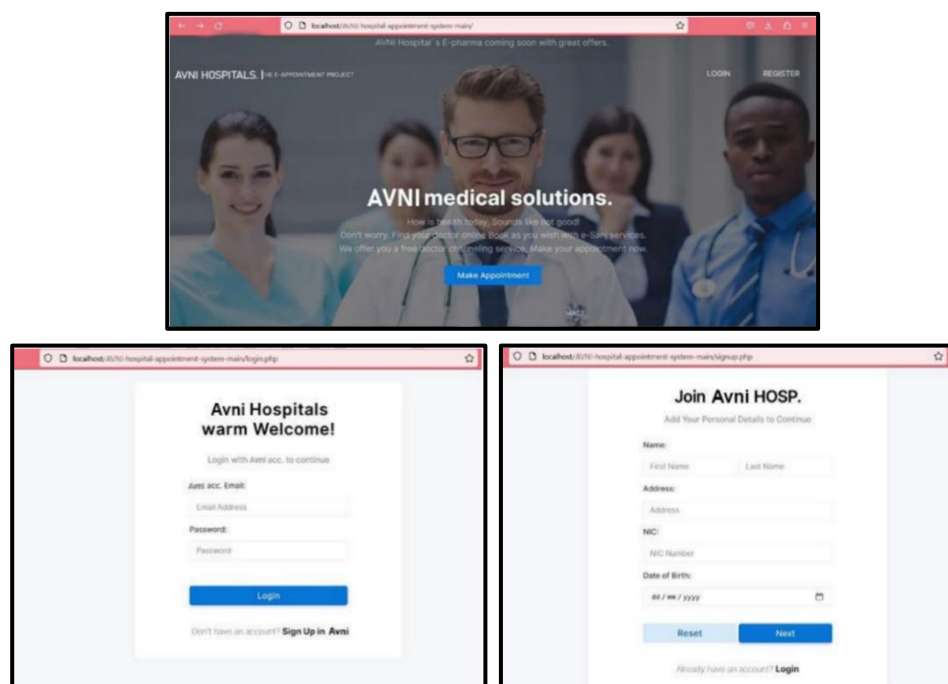


Figure 2.2 A Web-Based Doctor Appointment System

The study demonstrated the system's potential to significantly reduce waiting times, improve doctor-patient communication, and streamline hospital workflows. However, the research also noted that the system could be further enhanced by improving security features and integrating more advanced functionalities such as automated notifications and teleconsultation. Overall, the system represents a reliable and efficient solution for modern healthcare institutions seeking digital transformation (Srinivasan et al., 2024).

2.4.2. A Web-Based Online Doctor's Appointment and Medical Database Management System

This system was designed to replace the traditional manual medical appointment process with an automated online platform. The study, conducted at the Federal Polytechnic Ilaro Clinic, aimed to minimize patient waiting times and improve the management of medical records. The proposed web-based system allows patients to register, log in, and book appointments with available doctors based on their medical needs and location. The system architecture follows a three-tier design consisting of an interface, process management, and database management system. It was developed using technologies such as PHP, HTML5, JavaScript, and MySQL, following the Waterfall Model of system development. The platform includes three main user entities: the administrator, doctor, and patient. The administrator manages both patients and doctors, verifies appointments, and oversees database records; doctors can update patient details and approve or reject appointments; and patients can manage their personal records and view the status of their booked consultations.

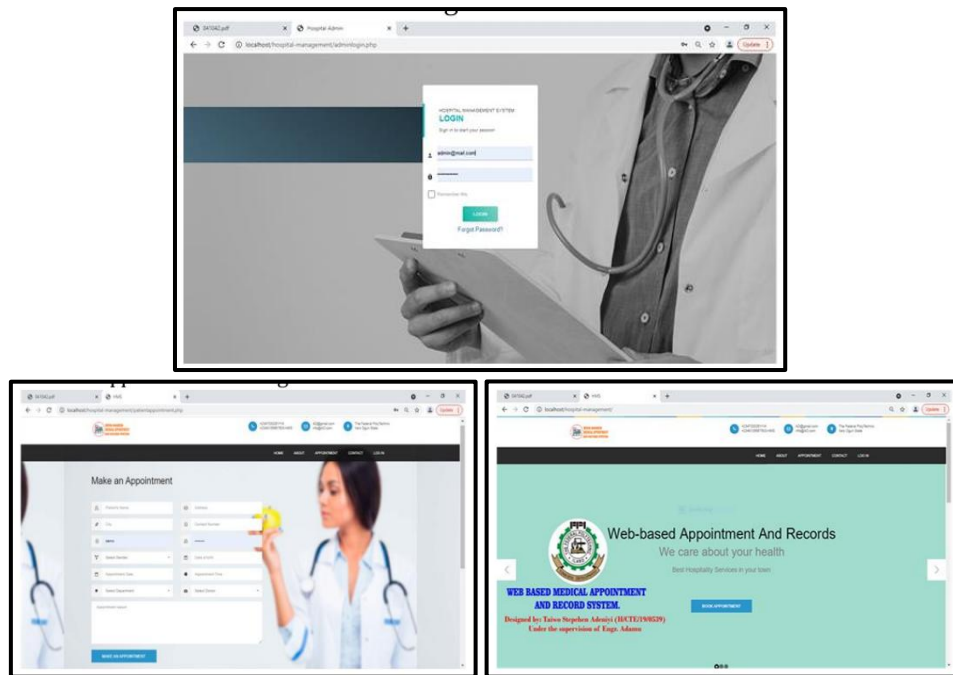


Figure 2.3 A Web-Based Online Doctor's Appointment and Medical Database Management System

The findings revealed that this system greatly enhances patient satisfaction by reducing overcrowding in clinics and eliminating the inefficiencies of manual booking. The researchers concluded that it is suitable for hospital deployment as it offers seamless appointment scheduling and secure patient data storage. Future recommendations include integrating automated notifications and improving database scalability for larger healthcare facilities (Akingbade & Onwuasoanya, 2022).

2.4.3. PEC e-SUS APS Online Appointment Scheduling System

This study focuses on the PEC e-SUS APS online appointment scheduling system developed in Brazil to enhance access to primary healthcare services and minimize patient waiting times. The system is integrated with the Conecte SUS Cidadão mobile application, allowing patients to schedule, view, and cancel appointments in real time through their smartphones. It also provides automated reminders and notifications to help reduce high no-show rates, a

common issue in Brazil's healthcare system. The PEC e-SUS APS platform operates alongside the national electronic health record, enabling health professionals to manage schedules, record appointments, and communicate updates efficiently.

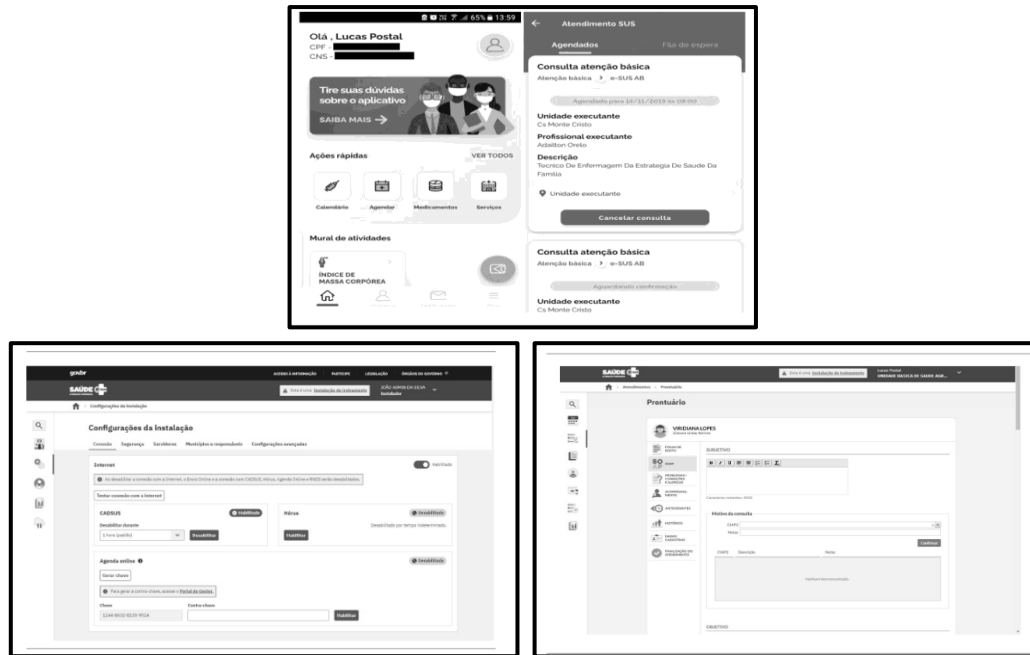


Figure 2.4 PEC e-SUS Aps Online Appointment Scheduling System

Despite its advanced integration with Brazil's Unified Health System (SUS), the study reveals that the adoption of the PEC e-SUS APS has been limited due to a lack of user training, technical awareness, and adherence to traditional appointment protocols. Health professionals also expressed concerns about losing control of scheduling processes and the risk of misuse by patients. However, the study highlights that, when properly implemented, the system can improve access to primary healthcare, reduce administrative workload, and promote patient satisfaction through convenient digital scheduling (Postal et al., 2021).

2.4.4. Smart Web-Based Medical Appointment Booking System (2022)

This study presents a smart web-based medical appointment booking system designed to replace traditional manual appointment methods and address common challenges in healthcare scheduling. The system enables patients and doctors to register, log in, and manage appointments through a secure online platform. Doctors can input their essential details, availability schedules, and areas of specialization, allowing the system to display real-time consultation slots. Patients, on the other hand, can conveniently select doctors, book available time slots, and receive appointment notifications. The system also stores patient medical records, giving both patients and doctors easy access to previous consultations and medical histories.

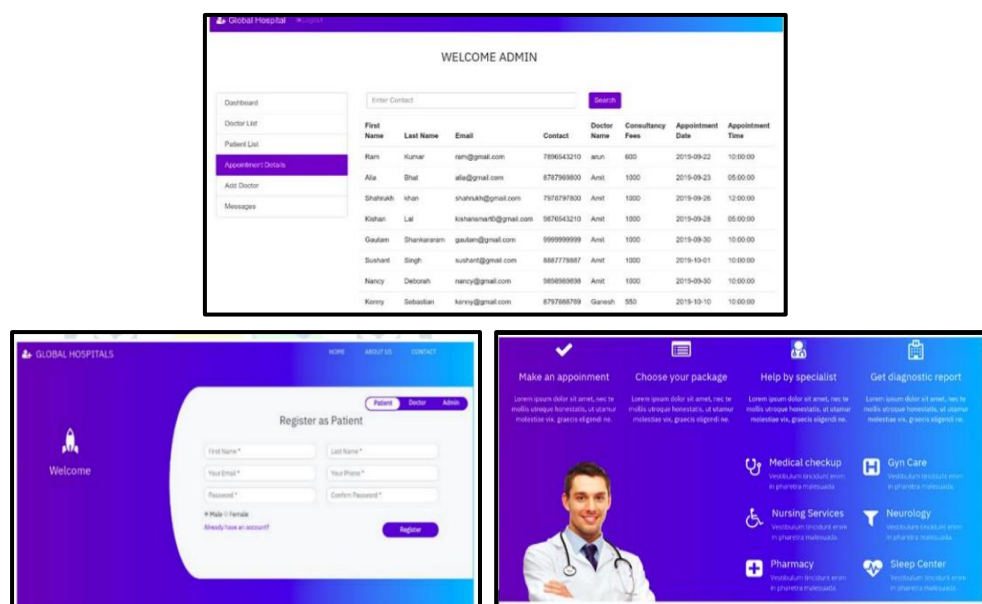


Figure 2.5 Smart Web-Based Medical Appointment Booking System

The system automates various processes that previously required phone calls or in-person visits, significantly reducing waiting times and improving the efficiency of hospital operations. It highlights the importance of accessible, well-organized doctor listings to assist users in quickly finding appropriate healthcare providers. Additionally, the study discusses related technologies such as cloud-

based appointment hosting, integrated reminders through SMS or email, and patient portals for secure communication. While the system effectively streamlines scheduling and enhances user convenience, the study notes that further improvements such as integrating more advanced features and optimizing user interaction could elevate its overall performance and adaptability in modern healthcare settings (JETIR, 2022).

2.5. Summary of the Related System

The related systems reviewed provided valuable insights that guided the development of this capstone project. The systems commonly offered online booking and schedule viewing, but most were designed only for desktop use and lacked mobile-optimized interfaces. Several important features were also missing, such as full doctor access, complete medical record integration, rescheduling functions, and consistent reminder notifications. By identifying these limitations, the proposed system aims to deliver a more complete and accessible solution for Mindalano Specialist Hospital, improving appointment handling, enhancing coordination among staff and doctors, and strengthening overall healthcare service delivery.

Table 2.1 Summary of Related Systems

Key Features	A Web-Based Doctor Appointment System	A Web-Based Online Doctor's Appointment and Medical Database Management System	PEC e-SUS APS Online Appointment System (Brazil)	Smart Web-Based Medical Appointment Booking System	Proposed System: Mindalano Specialist Hospital
Online Booking	✓	✓	✓	✓	✓
Reminders / Notifications	✗	✓	✓	✗	✓
Schedule Selection	✓	✓	✓	✓	✓
Reschedule Appointment	✗	✗	✓	✗	✓
Cancel Appointment	✗	✓	✓	✗	✓
Medical Records Access	✗	✓	✗	✗	✓
Doctor Access to System	✓	✓	✗	✓	✓
Mobile UI / Mobile Optimization	✗	✗	✗	✗	✓

2.6. Technical Background

The system is implemented as a **web-based application following a client–server architecture**, where the server is responsible for request processing, data management, and security enforcement, and clients access the system through web browsers. This architectural model enables centralized control of system resources and consistent data handling across all user roles.

The server-side application is developed using the **Django web framework**, which is based on the Python programming language. Django employs the **Model–View–Template (MVT)** architectural pattern, allowing clear separation between data models, application logic, and user interface templates. This structure supports modular development, maintainability, and controlled data access within the system.

Data storage and management are handled using a **relational database management system (MySQL)**. The database schema is designed to define relationships among entities such as users, appointments, schedules, and medical records. Constraints and relational keys are used to preserve data integrity and ensure consistent data transactions.

The presentation layer of the system utilizes **HTML, CSS, and JavaScript** to render dynamic web pages and handle client-side interactions. These technologies enable form validation, data display, and asynchronous user actions while maintaining compatibility across modern web browsers.

System security is implemented through **authentication and role-based authorization mechanisms**, ensuring that access to system functions and data is restricted according to user roles. Server-side validation and controlled database access are used to prevent unauthorized operations. The system also integrates **email services** to support automated notifications triggered by scheduling-related events.

This technical foundation defines the structural and operational components required for the development and execution of the system without addressing organizational context or implementation motivation.

.

2.6.1. Organizational Chart

The figure below shows the organizational chart of the Mindalano Specialist Hospital Foundation Inc.

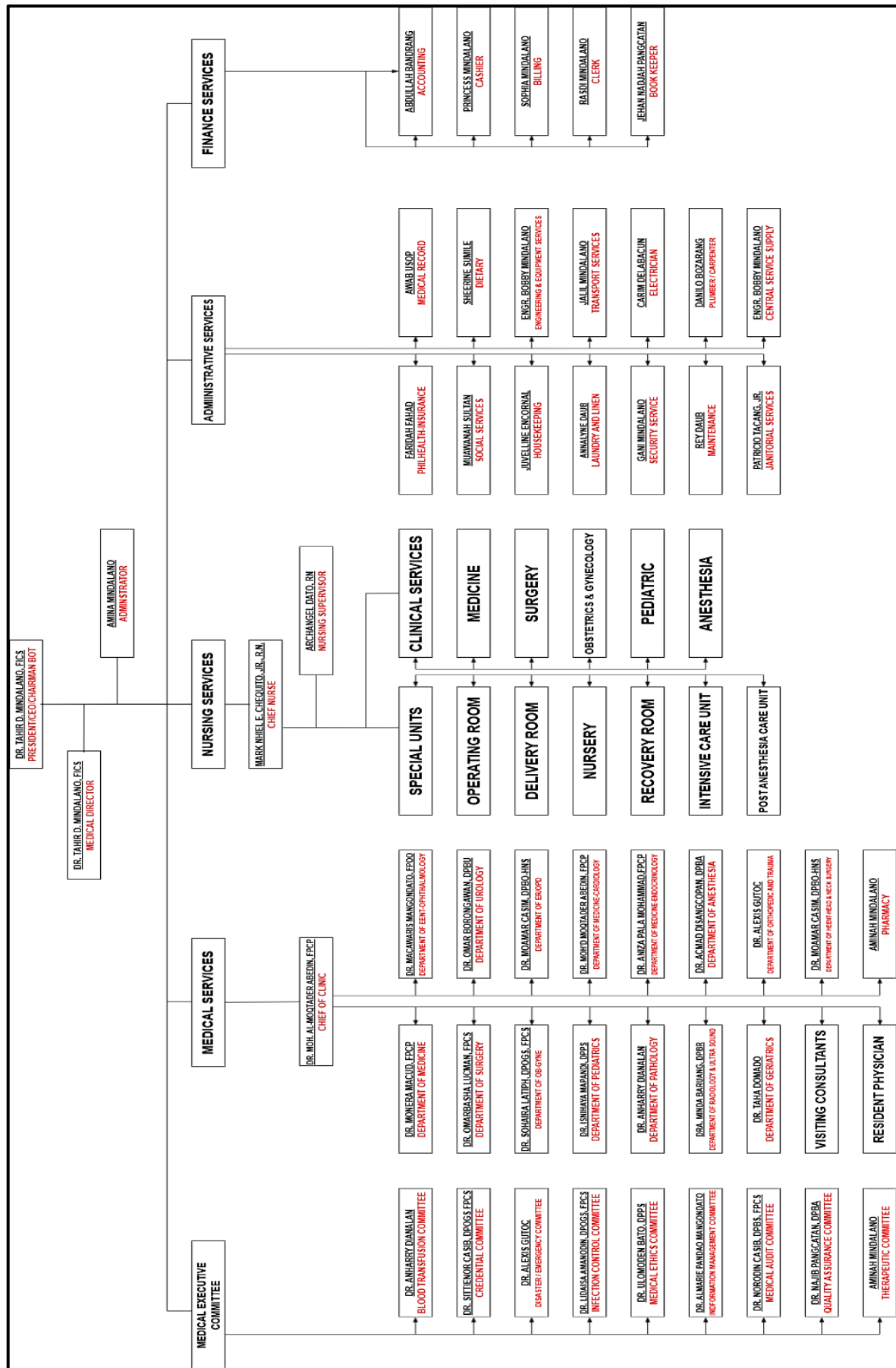


Figure 2.6 Organizational Chart

2.6.2. Workflow

This section presents the current and proposed workflows of Mindalano Specialist Hospital Foundation Inc., illustrating how patient registration, appointment processing, and consultation procedures are carried out. The workflow diagrams highlight the differences between the hospital's manual system and the improved processes introduced by the proposed web-based platform. By comparing the two workflows, this section demonstrates how the new system aims to reduce waiting times, minimize errors, enhance coordination, and improve the overall efficiency of hospital operations.

2.6.2.1. Current Workflow

This figure shows the existing workflow of the Mindalano Specialist Hospital Foundation Inc. (MHSFI).

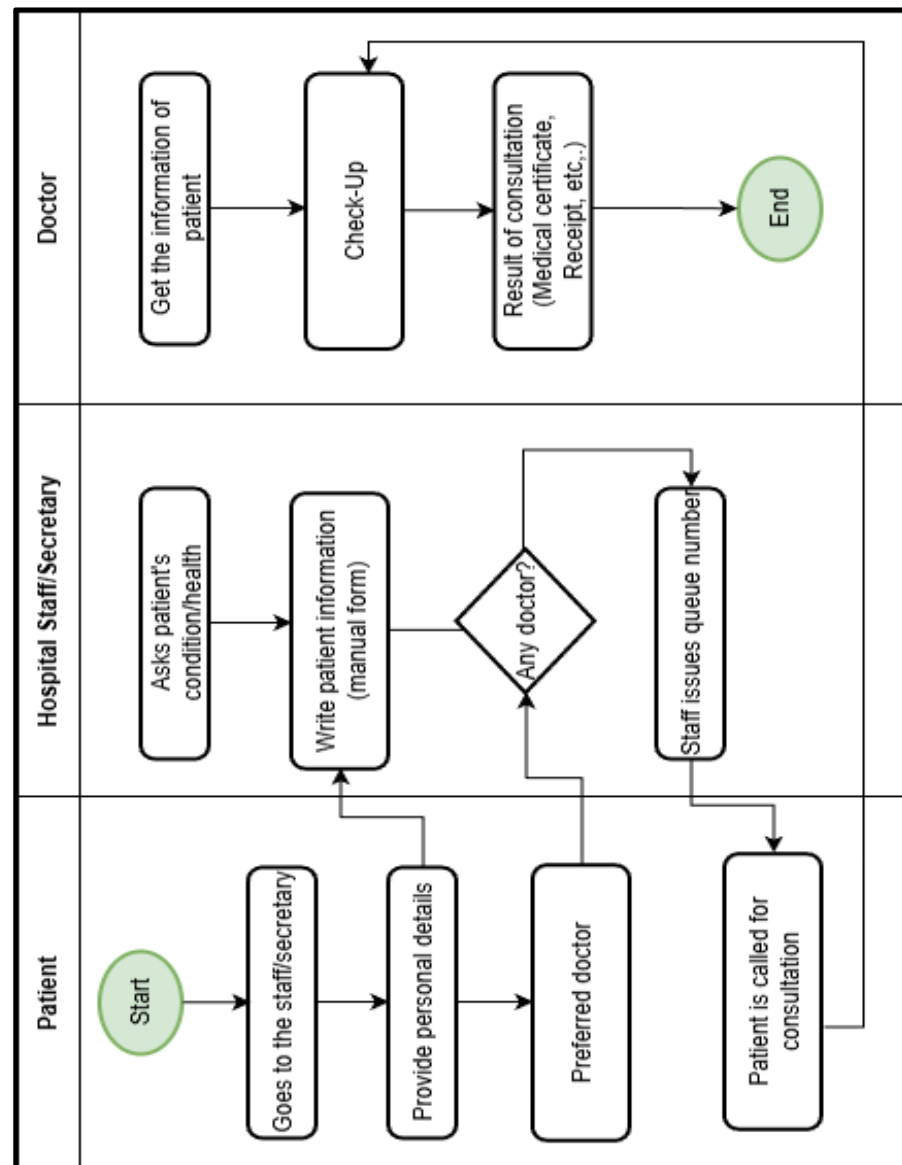


Figure 2.7 Current Workflow of MSHFI

The traditional workflow of Mindalano Specialist Hospital Foundation Inc. (MSHFI) is manual and paper-based process. Patients arrive at the hospital to obtain a priority number for consultation. They proceed to the staff or secretary, who conducts the initial assessment by asking about the patient's condition and recording basic information such as name, contact details, weight, blood pressure, and preferred doctor. All information is written manually on paper. After registration, the patient is issued a queue number and waits until they are called for consultation.

2.6.2.2. Proposed Workflow

This figure shows the proposed workflow of the Mindalano Specialist Hospital Foundation Inc. (MHSFI).

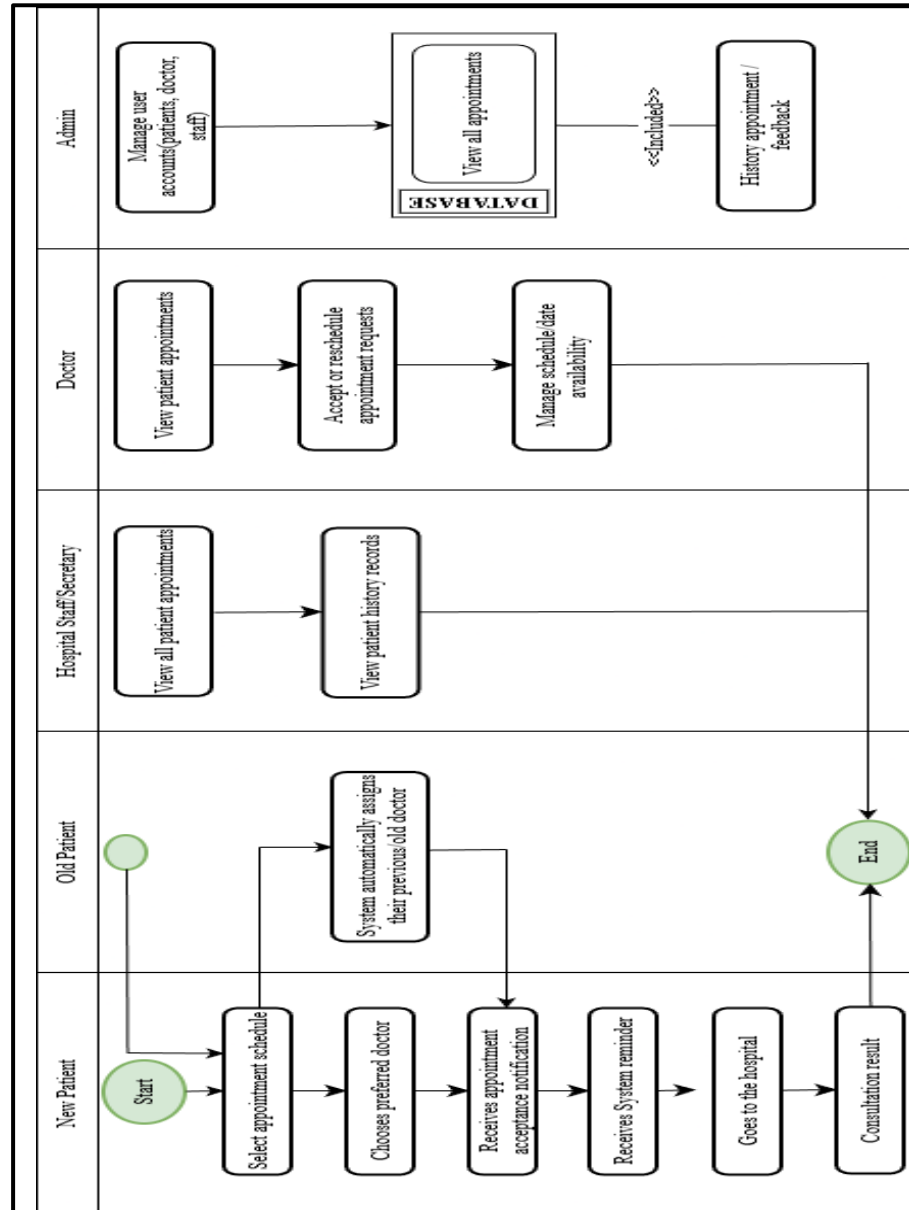


Figure 2.8 Proposed Workflow of MSHFI

The proposed workflow of Mindalano Specialist Hospital Foundation Inc. (MSHFI) shows how patients use the web-based system to book and reserve an appointment. After selecting their schedule and preferred doctor, the system notifies the patient once the appointment is accepted. On the day

of the consultation, the patient receives a system reminder for their scheduled check-up and then proceeds to the hospital for the consultation.

CHAPTER 3

METHODOLOGY

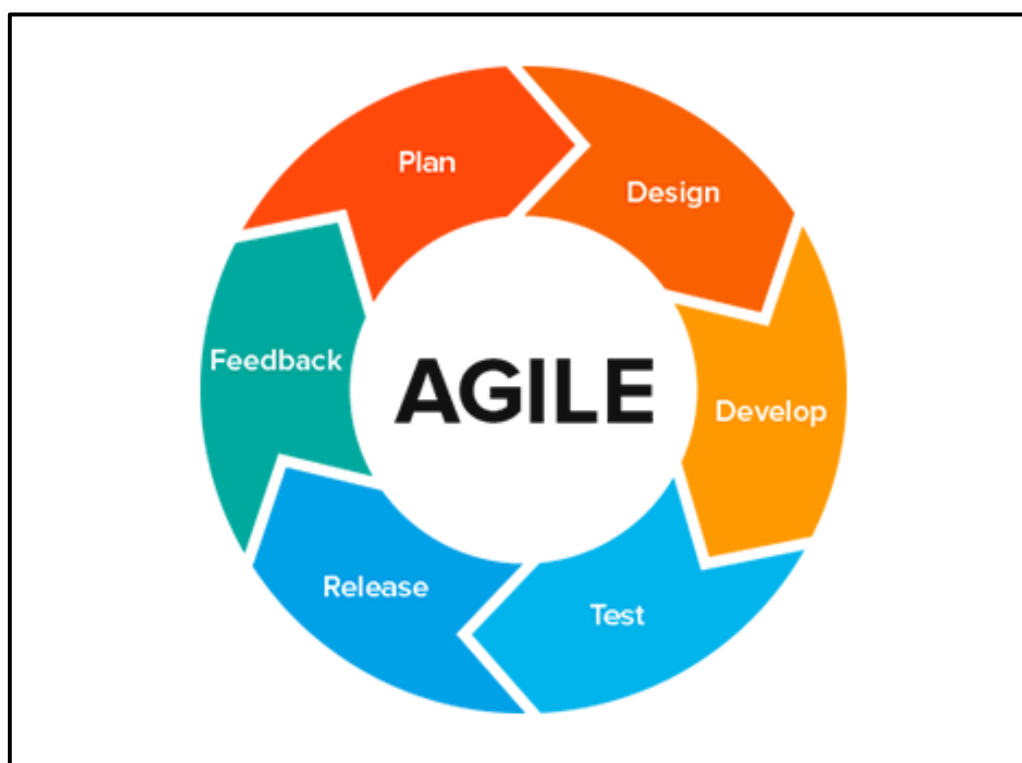


Figure 3.1 Agile Methodology

This chapter presents the methods and processes used in developing the Web-Based Medical Appointment and Scheduling System for Mindalano Specialist Hospital Foundation Inc. It describes the overall development framework, design models, system analysis procedures, and testing approaches applied throughout the project. By following a structured methodology, the proponents ensured that the system was systematically planned, iteratively developed, and thoroughly evaluated based on the needs of patients, doctors, staff, and administrators. The following sections explain the Agile approach used, the requirements identified, the system design diagrams, and the procedures involved in testing and validating the final system.

3.1. Scrum Framework

The development of the Web-Based Medical Appointment and Scheduling System followed the Scrum Framework, a lightweight Agile process that emphasizes iterative progress, collaboration, and continuous improvement. In this project, Scrum provided a structured yet flexible approach that allowed the proponents to break down the work into manageable sprint cycles. Each sprint focused on producing functional components such as user authentication, appointment booking, doctor schedule management, and notification modules that could be tested and evaluated immediately. The Product Owner, represented by hospital personnel, helped identify and prioritize the necessary system features, while the Scrum Master facilitated task organization and ensured adherence to Agile principles. The development team collaborated closely to design, code, and refine system modules based on collected requirements and user feedback.

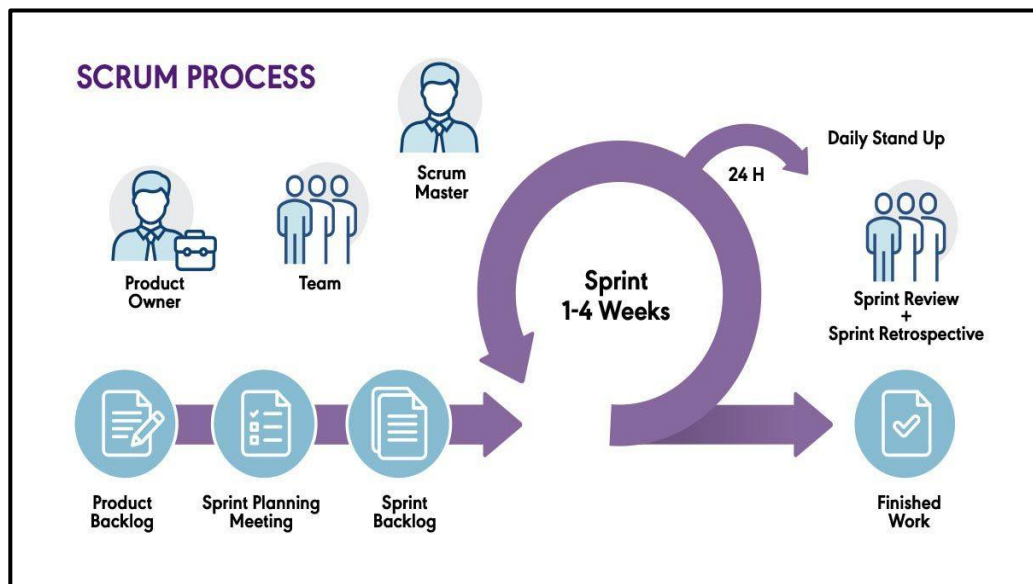


Figure 3.2 Scrum Framework

Scrum artifacts and events played a crucial role in ensuring clarity, transparency, and steady progress. The Product Backlog served as a repository of all identified features and improvements, while the Sprint Backlog outlined the

specific tasks to be completed within each cycle. At the end of every sprint, the team produced an Increment, which represented a working module ready for demonstration. Regular Sprint Reviews allowed hospital staff and stakeholders to evaluate the system's progress, provide feedback, and suggest adjustments based on real operational needs. Meanwhile, Sprint Retrospectives enabled the development team to assess their performance, identify challenges, and improve workflows for subsequent sprints. Through this cyclical and collaborative process, Scrum ensured that the system evolved effectively and aligned closely with the requirements and workflow of Mindalano Specialist Hospital Foundation Inc.

3.2. Requirements Analysis

This phase focused on identifying and understanding the essential requirements of stakeholders. The proponents conducted observations and informal interviews with staff members of Mindalano Specialist Hospital Foundation Inc., particularly those involved in managing appointments and patient scheduling. They were asked about the common issues encountered in the current manual process, such as long queues, scheduling conflicts, and delays in confirming appointments.

The proponents also gathered insights from individuals who had previously booked appointments or visited the hospital. Their feedback helped determine the necessary system features that would improve accessibility and convenience for patients. Through collaboration with hospital personnel and IT-related advisers, the team identified the functional and non-functional requirements of the system. These collected data became the foundation for the planning, design, and development of the medical appointment and scheduling platform.

3.2.1. Empathy Map

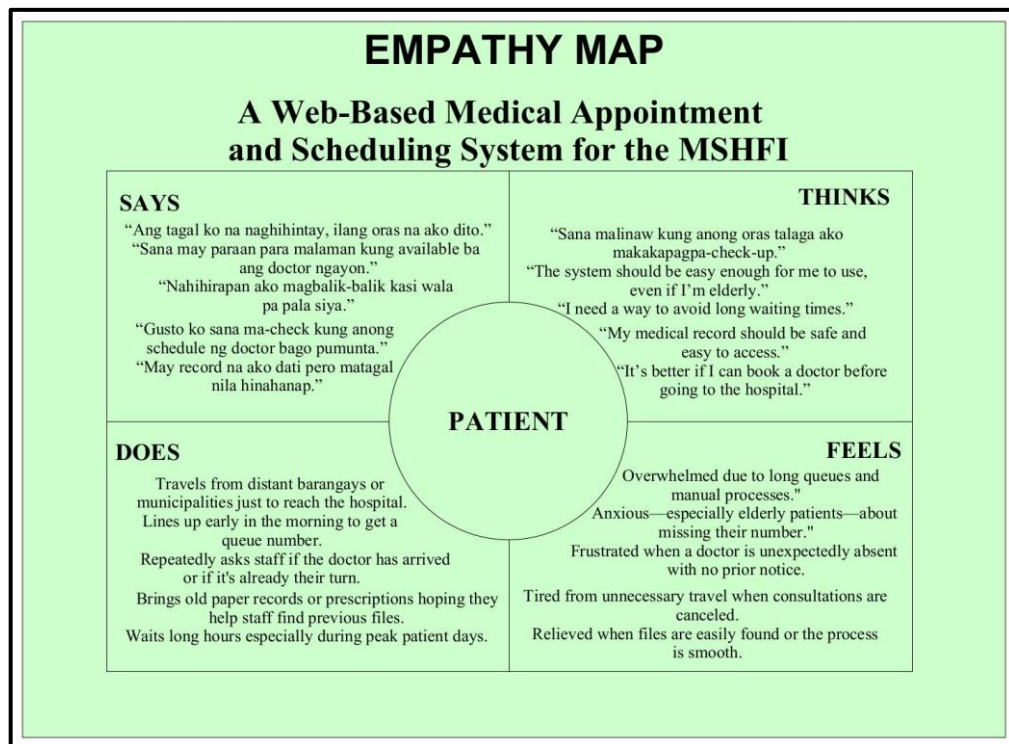


Figure 3.3 Empathy map of patient

The patient experiences long waiting times, uncertainty about doctor availability, and difficulty accessing medical records. They often travel far only to wait for hours, causing frustration and anxiety—especially among elderly patients. The patient needs a simple, reliable system to check schedules, book appointments in advance, and ensure their medical records are secure and easily accessible to reduce unnecessary delays and stress.

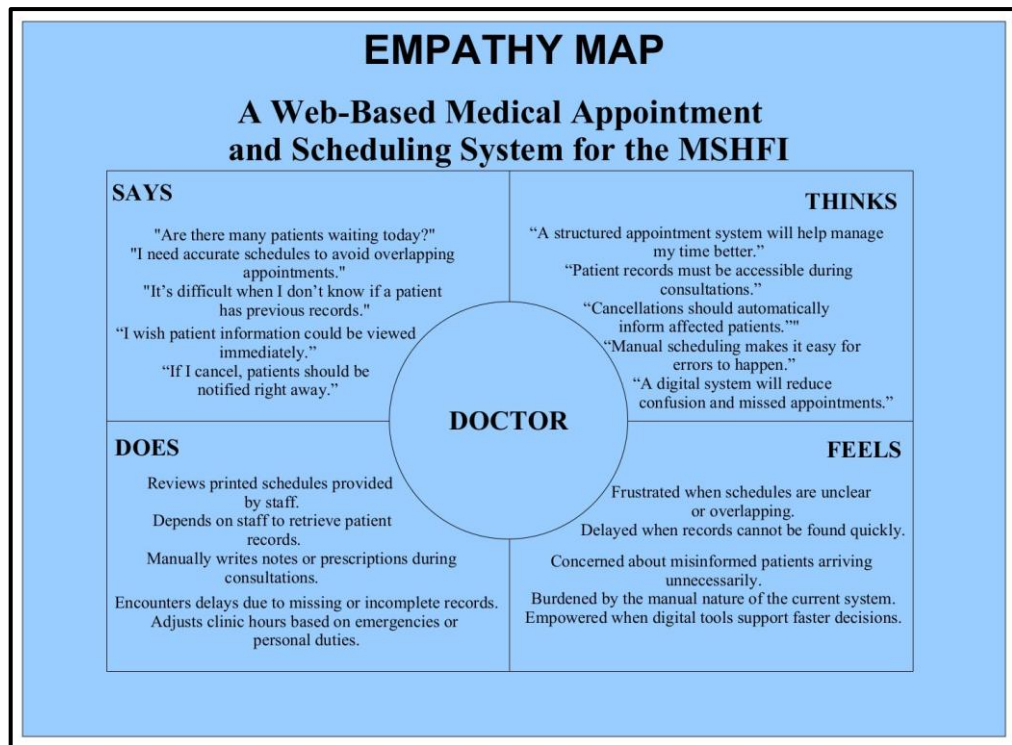


Figure 3.4 Empathy map of Doctor

The doctor relies on accurate schedules and complete patient records to provide effective consultations. Manual scheduling and missing records cause delays, confusion, and overlapping appointments. The doctor needs a structured, digital appointment and record system that provides real-time updates and automatic patient notifications, helping improve time management, reduce errors, and support better clinical decisions.

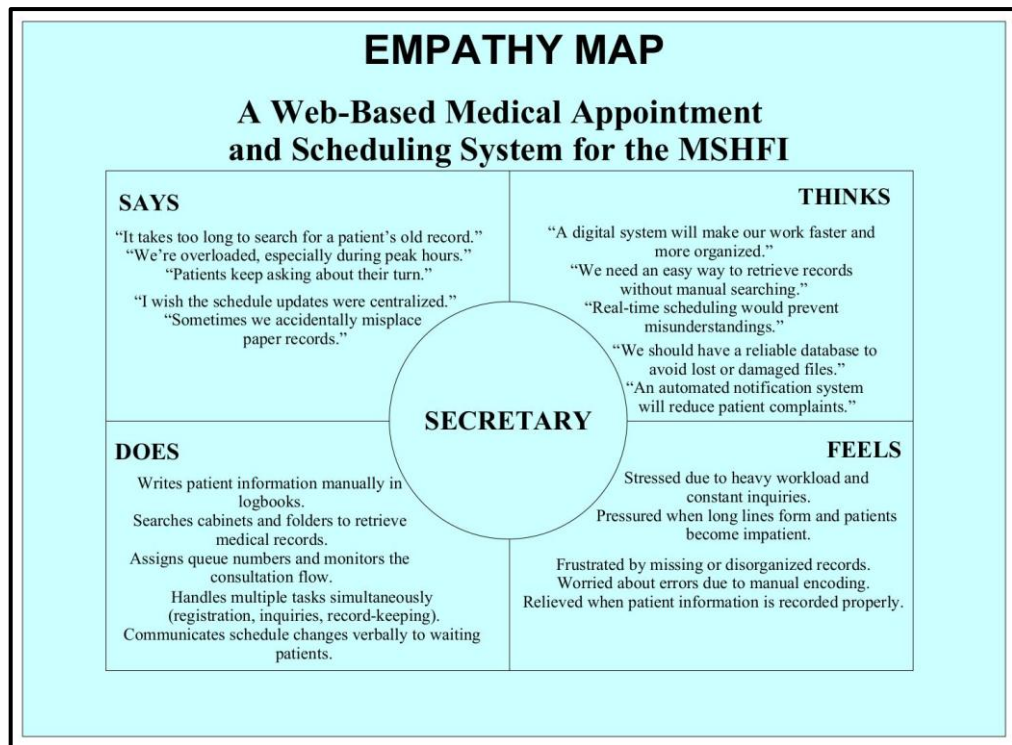


Figure 3.5 Empathy map of Secretary

The secretary handles heavy workloads, including manual record-keeping, scheduling, and constant patient inquiries. Searching for paper records and managing queues manually increases stress and the risk of errors. A centralized digital system is needed to organize schedules, retrieve records quickly, and automate notifications, allowing the secretary to work more efficiently and reduce patient complaints.

3.2.2. Functional and Non-Functional Requirements

Table 3.1 Functional and Non-Functional Requirements

REQUIREMENT ID	DESCRIPTION	TYPE	PRIORITY LEVEL
REQ-01	Patients register and log in securely.	Functional	First Priority
REQ-02	Patients book medical appointments.	Functional	First Priority
REQ-03	Patients modify or cancel bookings.	Functional	First Priority
REQ-04	Sends email alerts to patients.	Functional	First Priority
REQ-05	Doctors securely access their accounts.	Functional	First Priority
REQ-06	Doctors set consultation schedules.	Functional	First Priority
REQ-07	Doctors approve or decline appointment requests.	Functional	First Priority
REQ-08	Doctors view relevant patient information (Limited).	Functional	First Priority
REQ-10	Secretary log in using assigned accounts.	Functional	First Priority
REQ-11	Secretary view and update daily schedules.	Functional	First Priority
REQ-12	System Response Time: Respond within 2–4 seconds normally and not exceed 6 seconds under heavy usage.	Non-Functional	First Priority
REQ-13	Real-Time Processing: Database operations like booking, updating, and cancelling appointments must execute instantly.	Non-Functional	First Priority
REQ-14	System Uptime and Reliability: Must maintain 99% uptime with automated daily backups.	Non-Functional	First Priority
REQ-15	Security and Data Protection: Data must be protected through encryption and secure authentication (HTTPS, hashed passwords, role-based access control).	Non-Functional	First Priority
REQ-16	Usability: The system shall provide a simple and user-friendly interface that allows users to perform tasks efficiently with minimal training.	Non-Functional	First Priority
REQ-17	Patients update personal information.	Functional	Second Priority
REQ-18	Patients view past and upcoming visits.	Functional	Second Priority
REQ-19	Secretary encode walk-in patients.	Functional	Second Priority
REQ-20	Secretary assist patients with basic concerns.	Functional	Second Priority
REQ-21	Staff trigger reminders and updates (Notification Dispatch).	Functional	Second Priority
REQ-22	Secretary view appointment summaries (Limited Reporting Access)	Functional	Third Priority

REQ-23	Scalability: The system shall support an increasing number of users, appointments, and records without significant performance degradation.	Non-Functional	Third Priority
---------------	---	----------------	----------------

3.2.3. Gantt Chart

The Gantt Chart presents the project timeline, showing the sequence and duration of the major tasks involved in developing the Web-Based Medical Appointment and Scheduling System. It outlines the activities from requirement gathering, design, development, and testing up to documentation. This chart provides a clear overview of the project's progress and ensures that each phase is completed within the planned schedule.

Tasks / Activities	August	September	October	November	December	January	February	March	April	May	June
Project Planning and Requirements Gathering											
Literature Review											
Identify Stakeholders and Target Audience											
Survey Design and Preparation											
Data Collection from Stakeholders											
Data Analysis and Reporting											
Hierarchical Input-Process-Output & Input-Process-Output											
Use Case Diagram Development											
Entity-Relationship Diagram (ERD)											
Architectural Design											
Web Prototype											
Develop User and Admin Management System											
Implement User Authentication and Authorization											
Design User Interface (UI) for Web											
Implement Appointments and Scheduling Feature											
Integrate Security Features and Data Protection											
Testing											
Defense Preparation											
Documentation											

Table 3.2 Gantt Chart

3.3. Design

3.3.1. Hierarchical Input-Process-Output (HIPO)

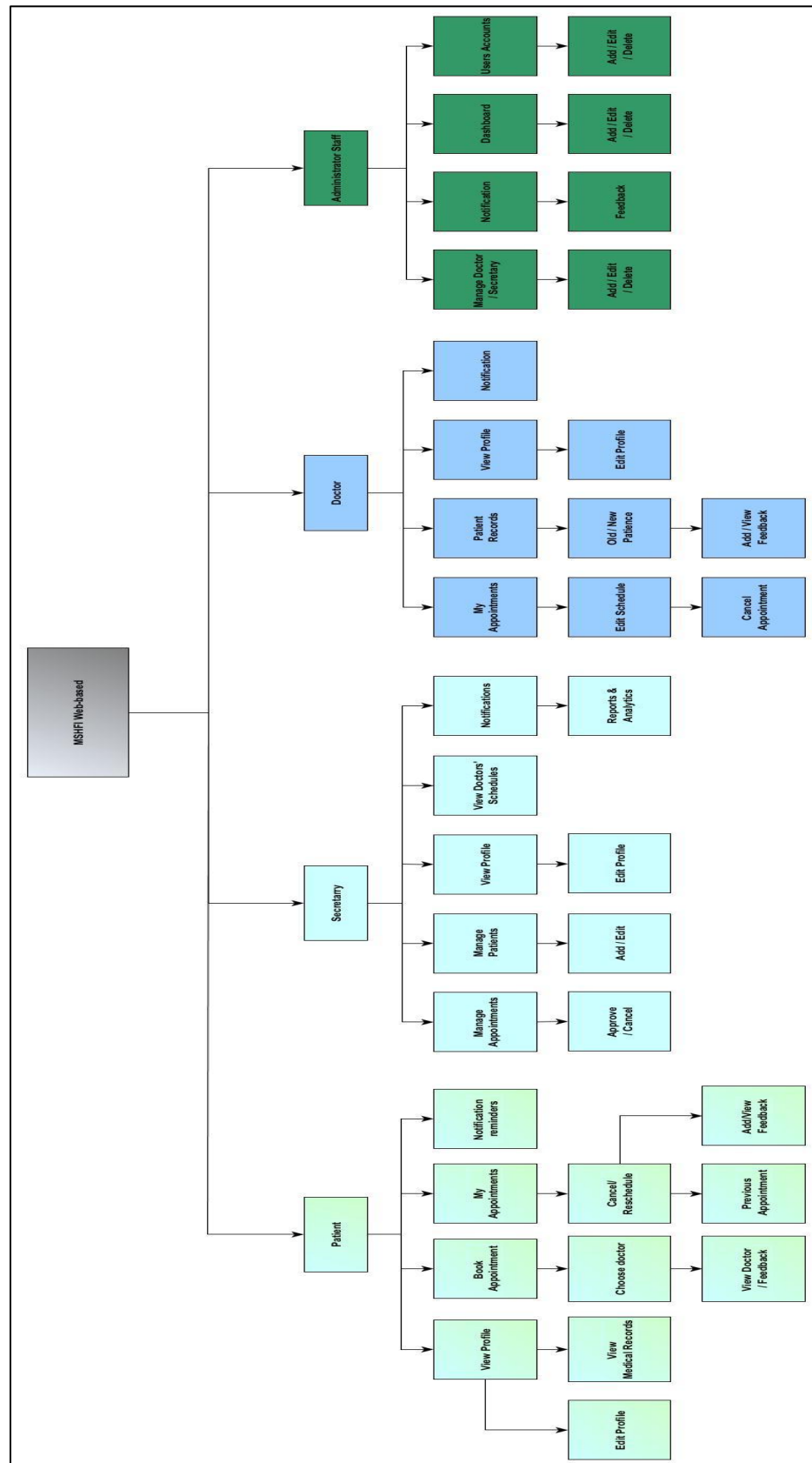


Figure 3.6 MSHFI Web-based Hierarchical Input-Process-Output

The MSHFI Web-based hierarchical input-process-output diagram serves as a visual roadmap, outlining the systematic flow of the process within the studied system. Therefore, the diagram summarizes the structured hierarchy of inputs, processes, and outputs, delivering a clear description of the complicated flow and process of the system.

3.3.2. Input-Process-Output (IPO) Diagram

3.3.2.1. Patient Input-Process-Output

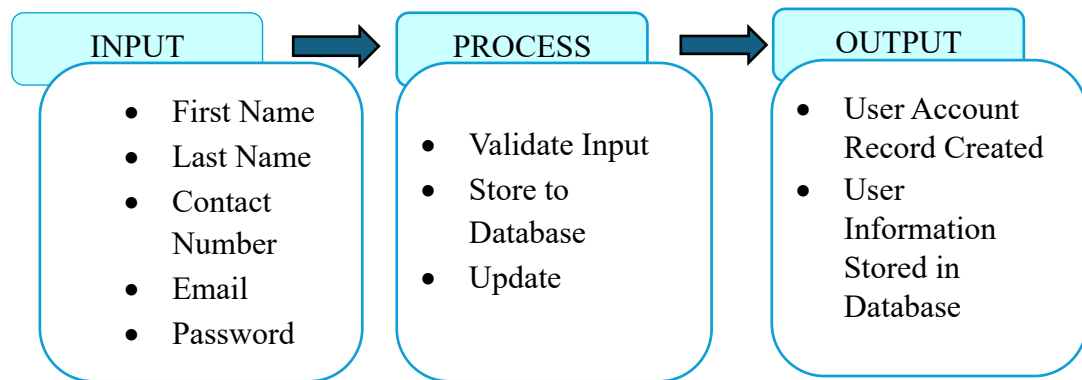


Figure 3.7 Input-Process-Output for User Account

This IPO diagram illustrates the patient account creation process, where the system collects basic patient information, validates the data, stores it in the database, and successfully creates a user account for future use.

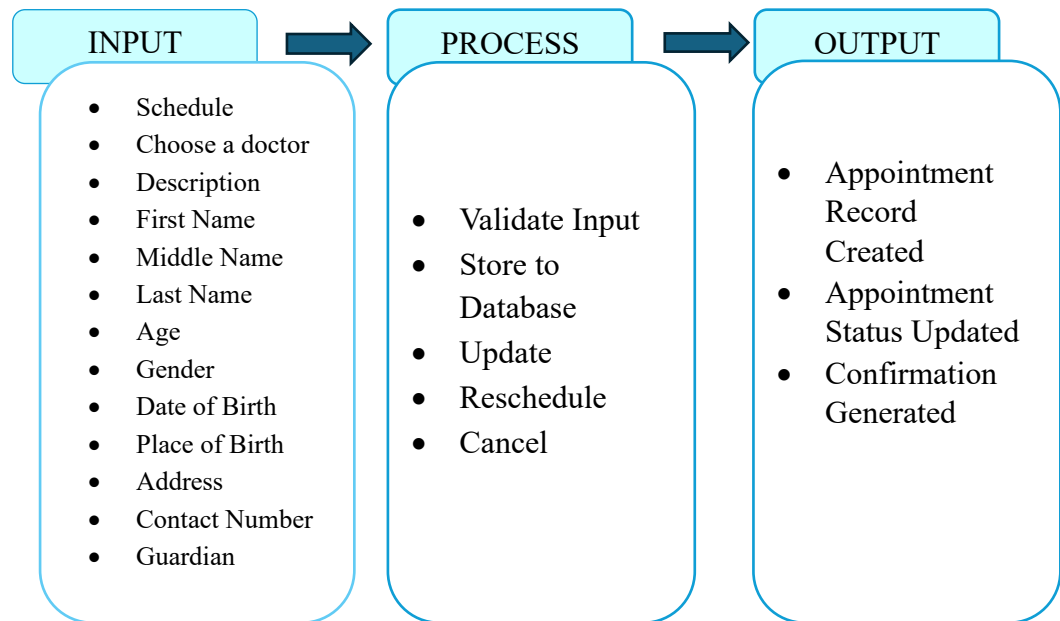
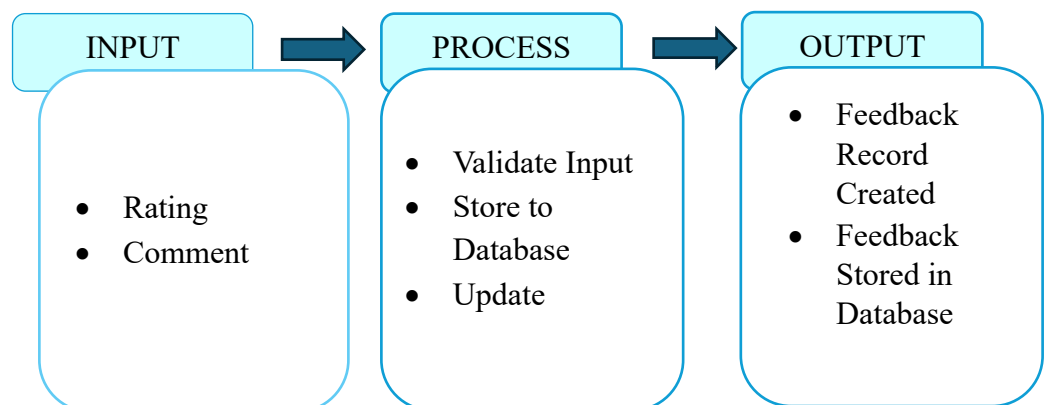


Figure 3.8 Input-Process-Output for Booking Appointment

This IPO diagram shows the appointment booking process, where the system collects patient and appointment details, validates and stores the information, manages updates or rescheduling, and generates an appointment confirmation.

Figure 3.9 Input-Process-Output for Feedback



This IPO diagram describes the feedback submission process, where the system collects patient ratings and comments, validates the input, and stores the feedback in the database for future review and evaluation.

3.3.2.2. Secretary Input-Process-Output

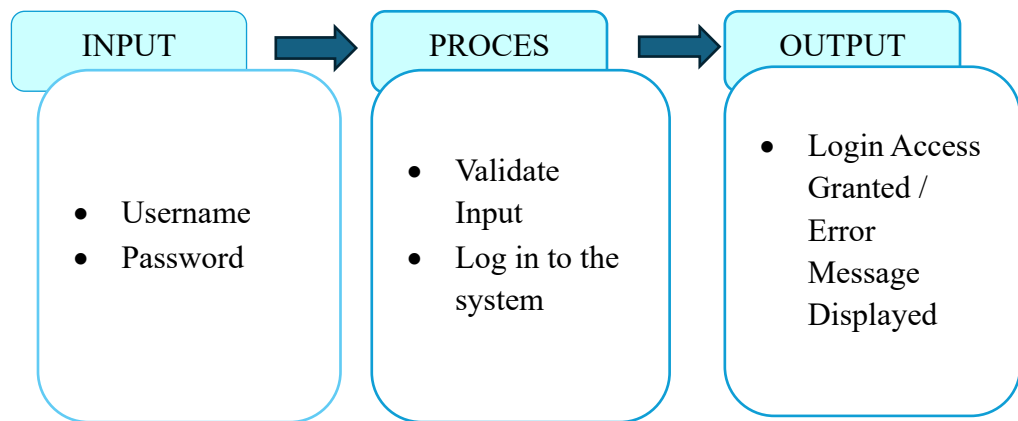


Figure 3.10 Input-Process-Output for Confirming Appointment

This IPO diagram shows the secretary login process, where the system validates the entered username and password and either grants access to manage appointments or displays an error message if authentication fails.

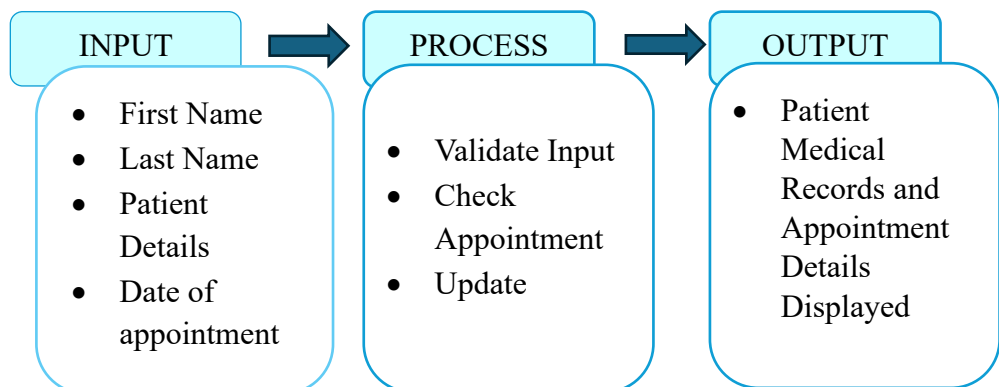


Figure 3.11 Input-Process-Output for View Medical Records

This IPO diagram describes the process of viewing patient medical records, where the system validates patient and appointment details, retrieves the relevant records from the database, and displays the information on the system interface.

3.3.2.3. Doctor Input-Process-Output

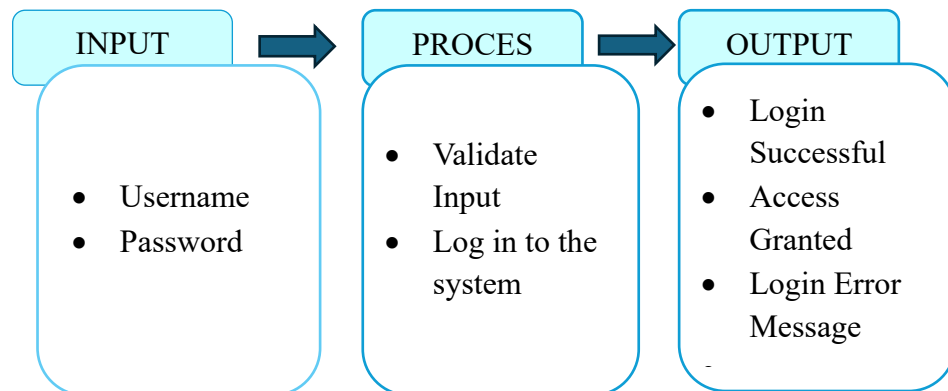


Figure 3.12 Input-Process-Output for User Account

This process enables the doctor to log in to the system using a valid username and password. The system verifies the entered credentials and either allows access upon successful validation or displays an error message if the login details are invalid.

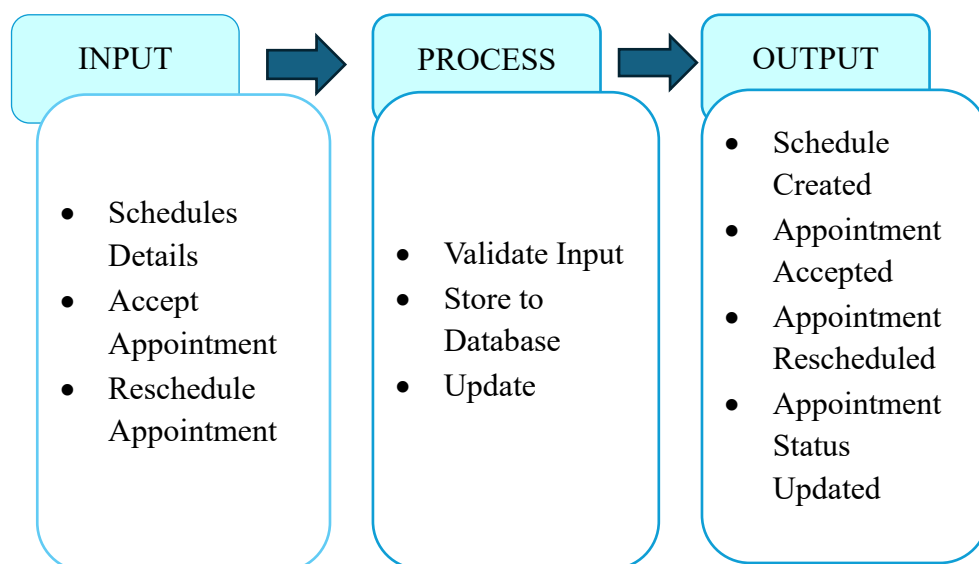


Figure 3.13 Input-Process-Output for Schedules and confirming/rescheduling an Appointment

This process allows the doctor to create schedules and manage appointments by accepting or rescheduling them. The system validates the

input, saves the changes to the database, and updates the appointment status accordingly.

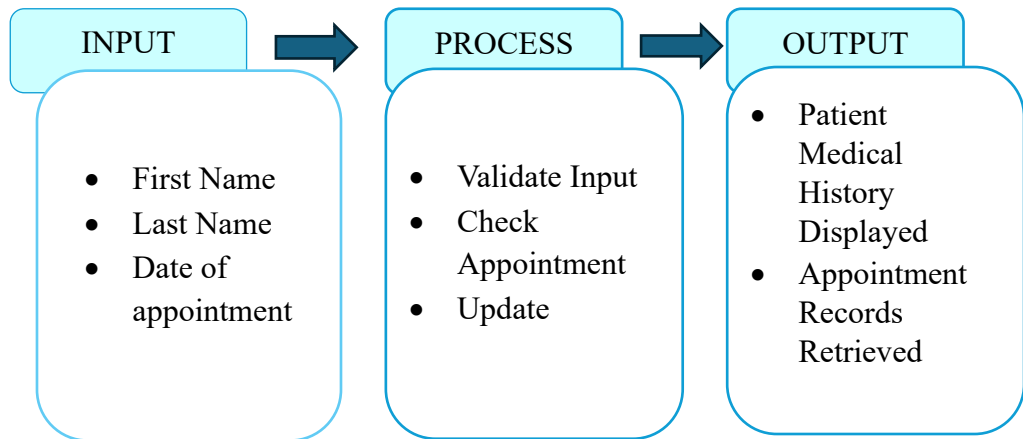


Figure 3.14 Input-Process-Output for viewing medical Records for a patient

This process allows the doctor to access a patient's medical records by entering basic patient and appointment details. The system validates the information, retrieves the relevant appointment data, and displays the patient's medical history.

3.3.2.4. Administrator Staff Input-Process-Output

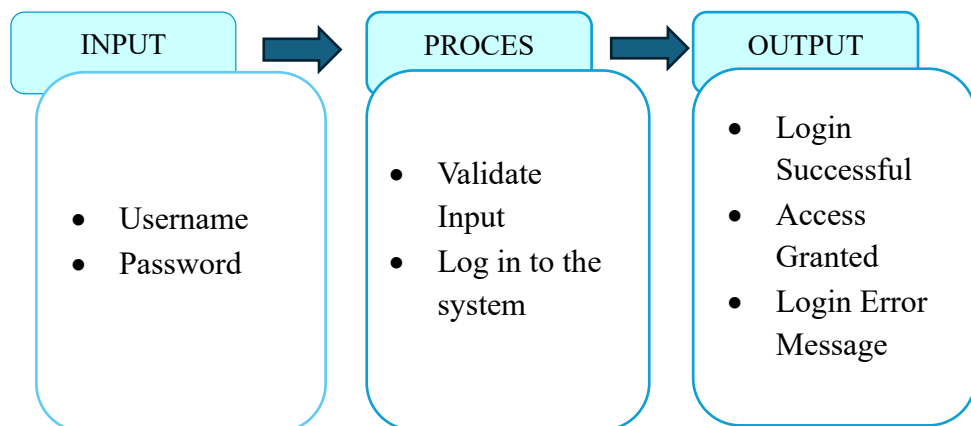


Figure 3.15 Input-Process-Output for User Account

This process enables the administrator staff to log in to the system using authorized credentials. The system verifies the username and password and either grants access upon successful authentication or shows an error message if the login attempt fails.

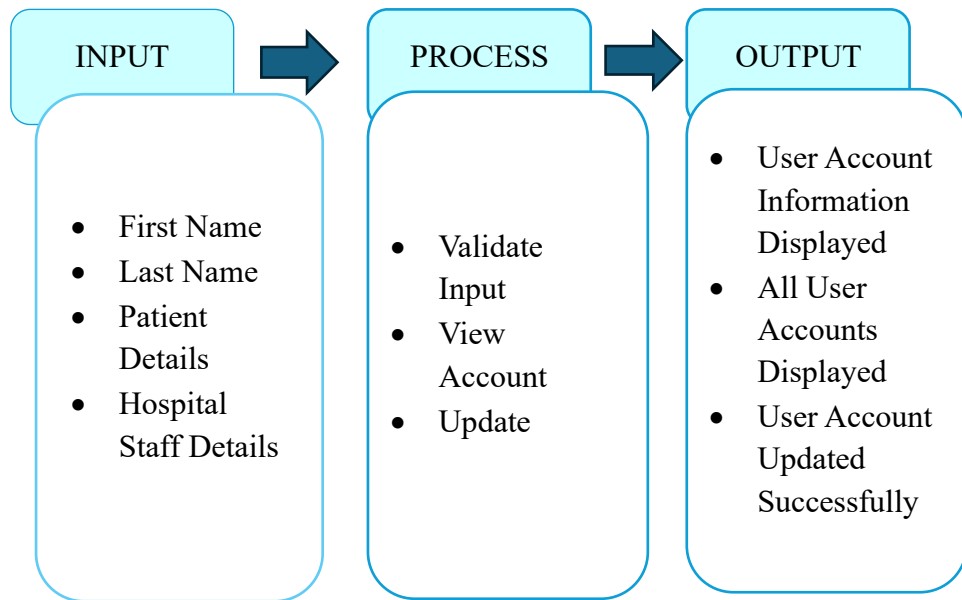


Figure 3.16 Input-Process-Output for Manage User Account

This process allows the administrator staff to view and manage user accounts by entering patient or hospital staff information. The system validates the input, retrieves the relevant account details, and updates user information when necessary.

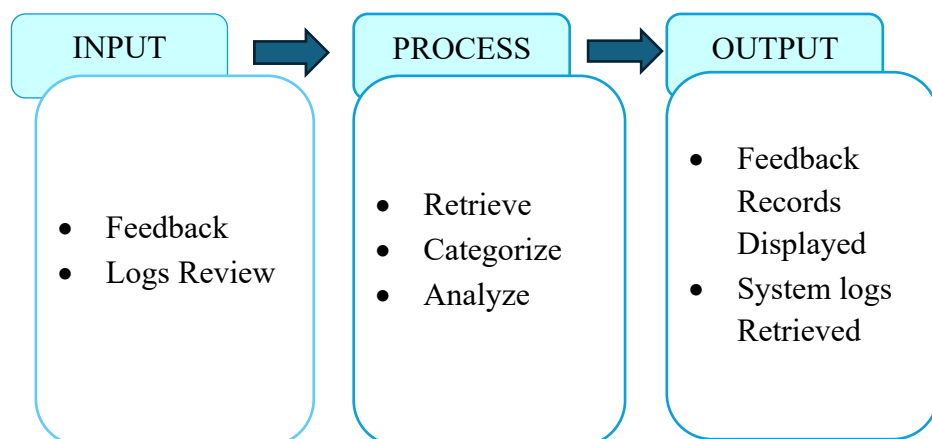


Figure 3.17 Input-Process-Output for Feedback & Logs Review

This process enables the administrator staff to review system feedback and activity logs. The system collects, organizes, and analyzes the records, then displays the feedback and log details for monitoring and evaluation purposes

3.3.3. Use Case Diagram

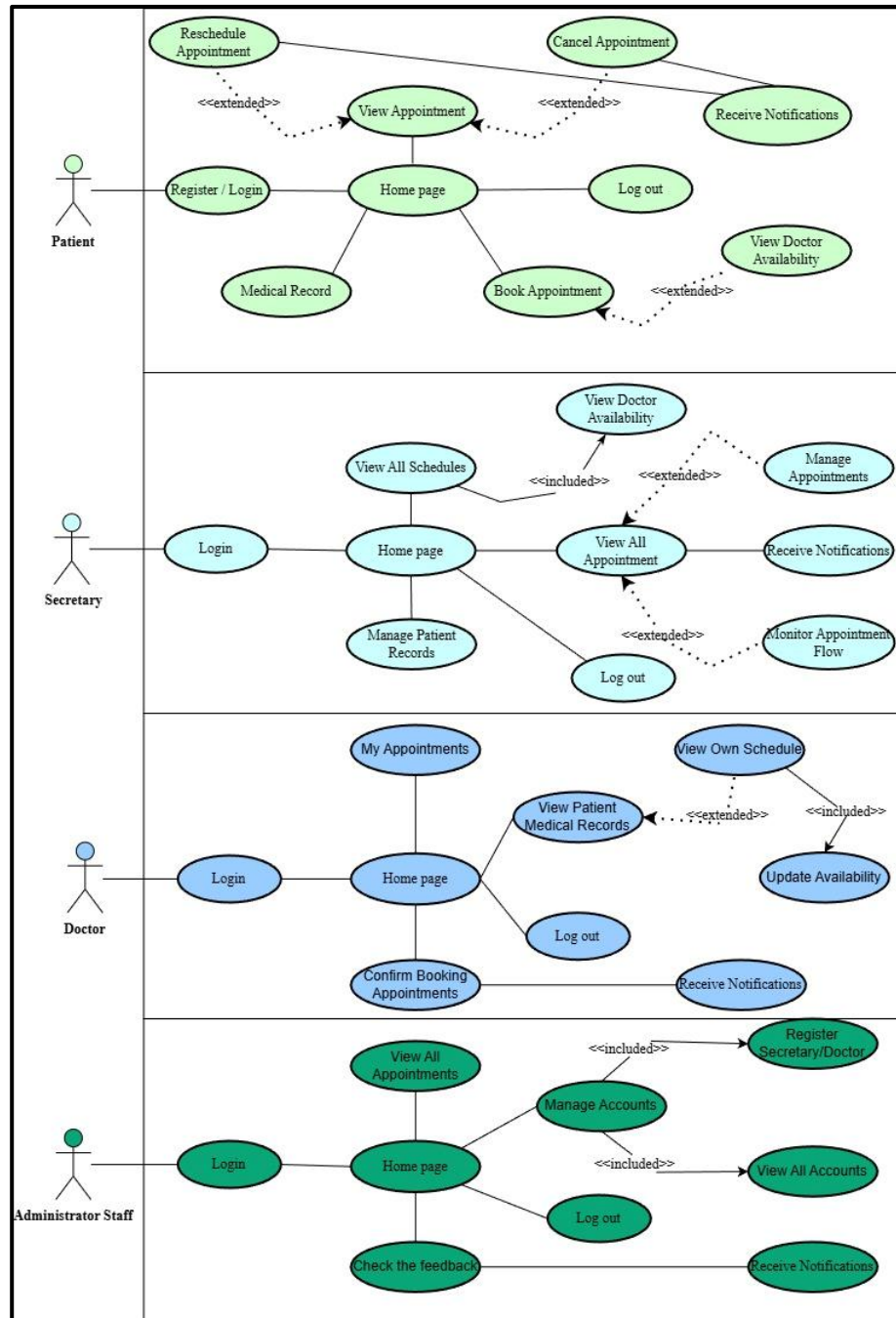


Figure 3.18 Use Case Diagram

Figure 3.18 shows the use case diagram of the A Web-Based Medical Appointment and Scheduling System for the MSHFI. It illustrates how different users interact with the medical appointment system. The system supports four main actors: Patient, Secretary, Doctor, and Administrator Staff, each with specific roles and responsibilities.

3.3.4. Entity-Relationship Diagram (ERD)

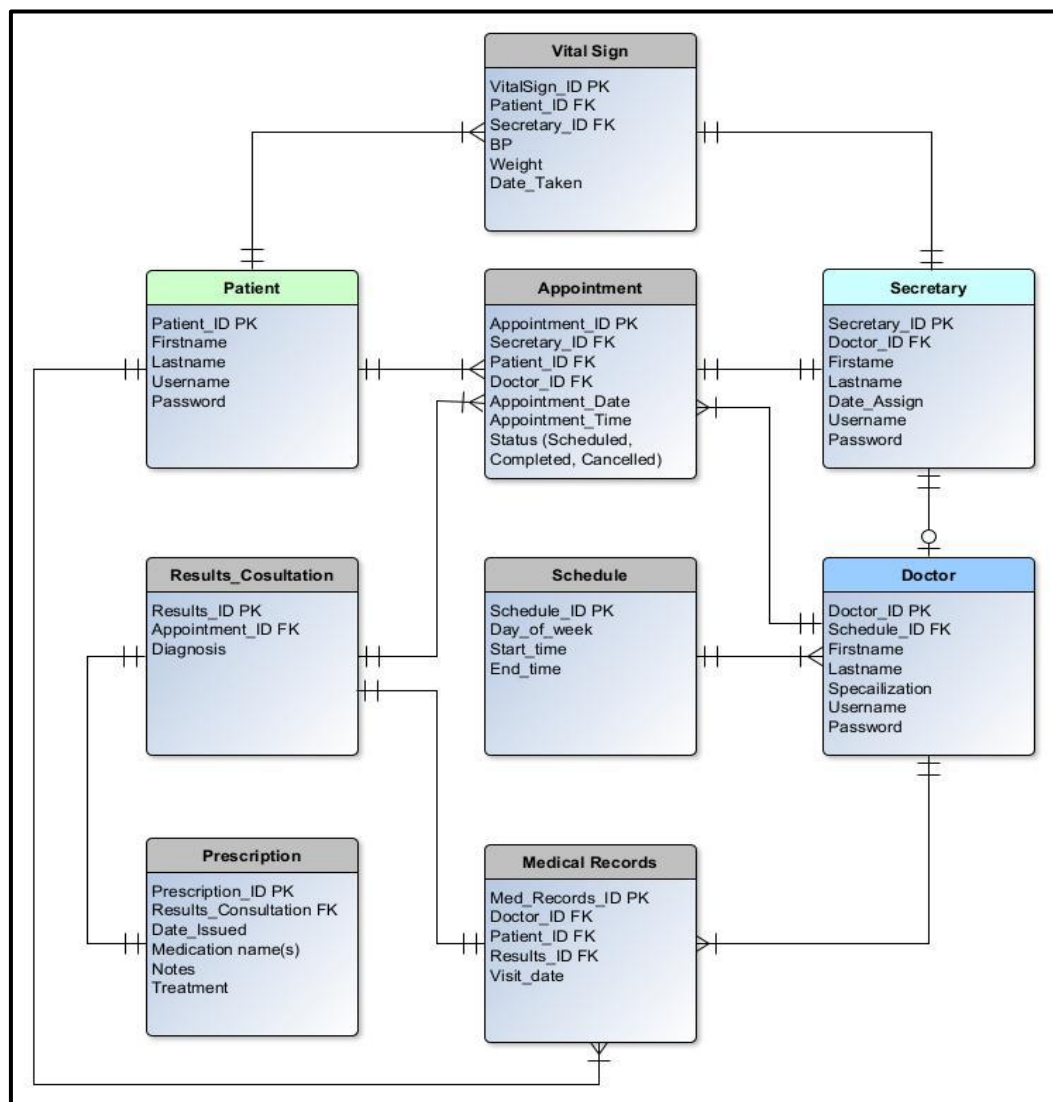


Figure 3.19 Entity-Relationship Diagram

The Entity Relationship Diagram represents the database structure of the Appointment Management System. It shows how patients, doctors, and secretaries are connected through appointments and schedules, while medical information such as vital signs, consultation results, prescriptions, and medical records are

stored and linked to each visit. The ERD ensures organized management of appointments and accurate recording of patient medical data.

3.3.5. Architectural Design

The Web-Based Medical Appointment and Scheduling System follows a client–server model implemented over the web. As shown in the figure below, users including patients, doctors, staff, and administrators access the system through personal computers, laptops, or mobile devices using a web browser. These client devices form the presentation layer, where users interact with the system, while the cloud infrastructure provides secure and reliable communication, ensures system availability, and routes requests to the application server.

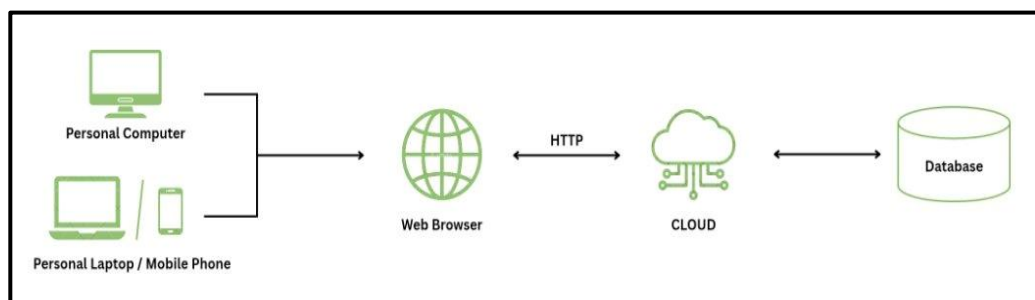


Figure 3.20 Architectural Design

The server hosts the core application logic, handling user authentication, appointment scheduling, doctor availability, notifications, and administrative functions. It processes requests according to business rules and validates transactions before interacting with the database, which stores user profiles, appointment records, doctor schedules, and activity logs. This separation of presentation, application, and data layers improves security, maintains consistency, and supports scalability, ensuring efficient performance and meeting the operational needs of Mindalano Specialist Hospital Foundation Inc.

3.4. Development

3.4.1. Software Requirements

Table 3.3 Developer's Software Requirements

Software	Specification
Operating System	Windows 10 or Windows 11
Programming Languages	JavaScript, HTML, CSS
Frameworks / Libraries	Tailwind, Django
Database	MySQL Xampp
Editor / IDE	Visual Studio Code
Additional Tools	Git / GitHub, Postman (API testing), Browser DevTools

Table 3.4 User's Software Requirements

Software	Specification
Registration / Authentication	Signup, Login, Email/Phone Verification (optional)
Booking	Select doctor, date, time, view availability
Notification	Email Alerts for appointment confirmation, cancellation, updates
Dashboard Access	Ability to view profile, appointments, and medical records (if enabled)

Table 3.5 Admin's Software Requirements

Software	Specification
Management	Add, Update, Delete doctors, schedules, services
Authentication	Admin login
Appointment Control	View appointments, approve/decline, manage status
Reports (optional)	Generate logs, statistics, patient visit history

Table 3.6 Server's Software Requirements

Software	Specification
Server Operating System	Linux (Ubuntu Server) or Windows Server
Web Server	Nginx or Apache
Application Runtime	Python
Web Framework	Django
Database Server	MySQL
Security Services	Firewall and SSL support

3.4.2. Hardware Requirements

Table 3.7 Developer's Hardware Requirements

System Component	Minimum	Recommended
Processor	Intel Core i3	Intel Core i7 or higher
RAM	4 GB	8–16 GB
Storage	500 GB HDD	1 TB HDD or 256 GB – 1 TB SSD
Internet Connection	5 Mbps	20 Mbps or higher (for API testing & cloud integration)
Keyboard & Mouse	Standard	Standard
Monitor	720p resolution	1080p or higher for UI development

Table 3.8 User's Hardware Requirements

System Component	Minimum	Recommended
Processor	Any Dual-Core Processor	Intel Core i3 or higher
RAM	2 GB	4–8 GB
Storage	200 MB free space (for browser caching)	500 MB free space
Display	720p resolution	1080p resolution
Internet Connection	3–5 Mbps	10 Mbps or higher

Input Devices	Standard Keyboard & Mouse / Touchscreen	Standard Keyboard & Mouse / Touchscreen
---------------	--	--

Explanation:

Users only need basic hardware because the system runs through a web browser. A stable internet connection ensures smooth booking, login, and notification access.

Table 3.9 Admin's Hardware Requirements

System Component	Minimum	Recommended
Processor	Intel Core i3	Intel Core i5 or i7
RAM	4 GB	8–16 GB (for handling multiple management pages)
Storage	500 GB HDD	1 TB HDD or 256 GB – 1 TB SSD
Display	1080p monitor	1080p or 1440p (for dashboard clarity)
Internet Connection	5 Mbps	20 Mbps or higher (for real-time updates & logs)
Input Devices	Standard Keyboard & Mouse	Standard Keyboard & Mouse

Table 3.10 Server Hardware Requirements

System Component	Minimum	Recommended
Processor	Quad-core processor	Intel Xeon or AMD EPYC
RAM	8 GB or higher	16 GB for better performance
Storage	256 GB SSD or higher	1 TB HDD or 256 GB – 1 TB SSD
Network	Stable broadband connection	Minimum 20 Mbps
Backup Storage	External or cloud-based backup	Automated daily backup

3.5. TESTING

3.5.1. Test Plan

The testers of the Web-Based Medical Appointment and Scheduling System for the Mindalano Specialist Hospital Foundation Inc. consist of three groups, each with distinct roles in evaluating the system. The first group is the researchers, who conduct internal testing to identify functional errors, interface inconsistencies, and potential usability issues. They ensure that the core features such as account creation, appointment booking, and notification systems operate correctly. The second group is composed of selected hospital staff, including administrative personnel and department heads, who assess the system's management functionalities. These staff members test administrative tasks such as managing doctor schedules, updating patient information, monitoring daily

quotas, and handling department workflows to ensure that the system aligns with real operational requirements. The third group consists of randomly selected outpatients who represent the actual end users of the system. These patients evaluate usability from the user perspective, checking the ease of navigation, clarity of instructions, responsiveness of the mobile application, and accuracy of notifications. By involving these three groups, the testing process captures a comprehensive understanding of system performance from technical, administrative, and user-centered viewpoints.

The testing procedure is designed to simulate real hospital operations while systematically evaluating the web . Testers are asked to perform a variety of tasks depending on their role, including creating and editing accounts, booking appointments, rescheduling or canceling reservations, navigating through menus and modules, managing doctor schedules, and verifying notifications. Each task is carefully observed, with researchers recording the testers' interactions, noting any errors, confusion, or difficulty experienced during the process. In addition to observational data, testers provide subjective feedback about usability, system clarity, and efficiency. The procedure also incorporates scenarios that test system limits, such as overlapping appointments, incomplete data entries, and full daily quotas, to evaluate robustness and error handling. This step-by-step approach ensures that the system is tested under realistic and challenging conditions, providing insights into both strengths and areas needing improvement.

The tools employed for testing support both functional validation and usability assessment of the Web-Based Medical Appointment and Scheduling System for the Mindalano Specialist Hospital Foundation Inc.. The System Usability Scale (SUS) questionnaire was used to gather structured feedback from

all participants, including outpatients, hospital staff and doctors, with ratings ranging from 1 (strongly disagree) to 5 (strongly agree). The SUS is a widely recognized tool for measuring usability and user satisfaction, providing quantitative feedback on dimensions such as learnability, efficiency, memorability, error tolerance, and overall satisfaction. Participants were asked to rate ten statements covering perceptions of system complexity, ease of use, need for technical support, integration of functions, consistency, confidence, and readiness for frequent use. In addition to the SUS, pre-designed test scripts guided participants consistently through each task to ensure all features were evaluated. Dummy data simulating patient profiles, appointments, and doctor schedules replicated real-world hospital operations, while browser developer tools was used to monitor performance, detect system errors, and verify responsiveness across devices. A local test server hosted the system for secure testing, and digital feedback forms collected testers' responses efficiently. Together, these tools enabled a systematic, thorough, and reliable evaluation of the system's functionality, usability, and user experience.

3.5.2. Respondents

The respondents of the study were selected to evaluate the usability, effectiveness, and efficiency of the Web-Based Medical Appointment and Scheduling System. These respondents include patients, hospital staff, and doctors of Mindalano Specialist Hospital Foundation Inc. To determine the appropriate number of respondents needed for the evaluation, the proponents used Slovin's Formula, which is commonly applied when the total population is known but variability within the population is not.

This study utilized simple random sampling as the sampling technique. Simple random sampling is a probability sampling method in which every member of the population has an equal and non-zero chance of being selected as a respondent. This method was used to ensure that the selection of participants was unbiased and representative of the entire population. The respondents were chosen randomly using a random selection process such as a random number generator. Each member of the population was given an equal opportunity to be included in the study, which minimized researcher bias and enhanced the reliability and validity of the research findings. This sampling technique was deemed appropriate for the study as it provides fair representation and supports the generalization of results.

3.5.3. Tools

System Usability Scale (SUS) Questionnaire

Participants rated the following statements using a five-point Likert scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree.

Table 3.11 System Usability Scale (SUS) Questionnaire

No.	STATEMENT	1	2	3	4	5
1.	I think that I would like to use this system frequently.					
2.	I found the system unnecessarily complex.					
3.	I thought the system was easy to use.					
4.	I think that I would need the support of a technical person to be able to use this system.					
5.	I found the various functions in this system were well integrated.					

6.	I thought there was too much inconsistency in this system.					
7.	I would imagine that most people would learn to use this system very quickly.					
8.	I found the system very cumbersome to use.					
9.	I felt very confident using the system.					
10.	I needed to learn a lot of things before I could get going with this system.					

REFERENCES

- [1] AbouZahr, C., & Boerma, T. (2005). Health information systems: the foundations of public health. *Bulletin of the World Health Organization*, 83, 578–583.
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2626318/>
- [2] Aljabri, D., et al. (2023). Impact of automated notification systems on patient engagement and satisfaction in healthcare: A systematic review. *Journal of Medical Systems*, 47(3), 112.<https://pmc.ncbi.nlm.nih.gov/articles/PMC9940665/>
- [3] Aljabri, M., Althobaiti, A., & Alsubaie, A. (2023). Impact of automated notification and reminder systems on patient engagement and appointment compliance. *PMC*.
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9893353/>
- [4] A Web-Based Doctor Appointment System. *ResearchGate*.
https://www.researchgate.net/publication/385379804_A_WEB-BASED_DOCTOR_APPOINTMENT_SYSTEM
- [5] A Web-Based Doctor Appointment System. *International Research Journal of Engineering and Technology (IRE Journals)*.
<https://www.irejournals.com/paper-details/1703874>
- [6] Blumenthal, D., & Tavenner, M. (2021). The meaningful use regulation and electronic health records. *New England Journal of Medicine*, 385(4), 301–303.
- [7] Chung, S. (2020). Patient-Centric Scheduling With the Implementation of a Web-Based Appointment System. *Journal of Medical Internet Research*, 22(6), e16451.
<https://www.jmir.org/2020/6/e16451/>
- [8] Chung, K. (2020). Patient-centric web-based scheduling systems: Improving access and patient experience in healthcare. *PMC*.
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7315363/>

- [9] De Carvalho Junior, M. A., & Bandiera-Paiva, P. (2018). Health Information System Role-Based Access Control: Current Security Trends and Challenges. *Journal of Healthcare Engineering*, 2018. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5836325/>
- [10] Edgman-Levitan, S., & Schoenbaum, S. C. (2021). Patient-centered care: achieving higher quality by designing care through the patient's eyes. *Israel Journal of Health Policy Research*, 10, 21. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7934513/>
- [11] Epizitone, A. (2023). A systematic literature review of health information systems. *Healthcare (MDPI)*, 11(7), 959. <https://www.mdpi.com/2227-9032/11/7/959>
- [12] Epizitone, A. I. (2023). Adoption and impact of electronic medical records in community and outpatient healthcare: A systematic review. *PMC*. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9893353/>
- [13] *Journal of Emerging Technologies and Innovative Research (JETIR)*. (n.d.). <https://www.jetir.org/>
- [14] Lin, Y., Chen, H., & Zhang, T. (2022). Automated patient self-scheduling in healthcare systems: Effects on operational efficiency and patient satisfaction. *PMC*. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9382371/>
- [15] *Medicine*. (2023). Effectiveness of internet-based appointment systems in healthcare: A comparative study with manual methods. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMCXXXXXX/> (replace XXXXX with actual article ID if needed)
- [16] National Center for Biotechnology Information. (2023). Article PMC9940665. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9940665/>
- [17] National Library of Medicine. (2021). PubMed Article 34231716. <https://pubmed.ncbi.nlm.nih.gov/34231716/>

- [18] Philippine Statistics Authority (2020). Population and Housing Census: Lanao del Sur.
<https://psa.gov.ph/content/2020-census-population-and-housing-results>
- [19] Zhao, P., et al. (2017). Web-Based Medical Appointment Systems: A Systematic Review. Journal of Medical Internet Research, 19(4), e134.
<https://www.jmir.org/2017/4/e134/>